



Traffic Mirroring Configuration Guide for Cisco 8000 Series Routers, Cisco IOS XR Releases

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Preface

This cumulative guide provides a single, continuously updated version that includes all the latest IOS XR features and release updates. It simplifies your experience by letting you bookmark one link and access the complete guide, instead of navigating through multiple release-specific versions.

Specific changes or updates tied to individual releases are clearly called out within the relevant sections. For a list of features introduced in a specific release, refer to the [Release Notes](#) or the [IOS XR Feature Finder](#).

The table lists the release numbers for which this document has been updated since its initial publication.

Table 1: Changes to this document

Date	Summary
December 2025	Republished for Release 25.4.1
October 2025	First published for Release 25.3.1

Communications, Services, and Additional Information

- To receive timely, relevant information from Cisco, sign up at [Cisco Profile Manager](#).
- To get the business results you're looking for with the technologies that matter, visit [Cisco Services](#).
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- To obtain general networking, training, and certification titles, visit [Cisco Press](#).
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Cisco Bug Search Tool

[Cisco Bug Search Tool](#) (BST) is a web-based tool that acts as a gateway to the Cisco bug tracking system that maintains a comprehensive list of defects and vulnerabilities in Cisco products and software. BST provides you with detailed defect information about your products and software.



CHAPTER 1

YANG Data Models for Traffic Mirroring

This chapter provides information about the YANG data models for traffic mirroring features.

- [Using YANG Data Models, on page 1](#)

Using YANG Data Models

Cisco IOS XR supports a programmatic way of configuring and collecting operational data of a network device using YANG data models. Although configurations using CLIs are easier and human-readable, automating the configuration using model-driven programmability results in scalability.

The data models are available in the release image, and are also published in the [Github](#) repository. Navigate to the release folder of interest to view the list of supported data models and their definitions. Each data model defines a complete and cohesive model, or augments an existing data model with additional XPath. To view a comprehensive list of the data models supported in a release, navigate to the *Available-Content.md* file in the repository.

You can also view the data model definitions using the [YANG Data Models Navigator](#) tool. This GUI-based and easy-to-use tool helps you explore the nuances of the data model and view the dependencies between various containers in the model. You can view the list of models supported across Cisco IOS XR releases and platforms, locate a specific model, view the containers and their respective lists, leaves, and leaf lists presented visually in a tree structure. This visual tree form helps you get insights into nodes that can help you automate your network.

To get started with using the data models, see the *Programmability Configuration Guide*.



CHAPTER 2

Getting Started with Traffic Mirroring

This chapter introduces traffic mirroring, key concepts, traffic mirroring types, and key terminology. Users can learn about how traffic mirroring works to enable network analysis without impacting live traffic.

- [Traffic mirroring, on page 3](#)
- [Configuration guidelines for traffic mirroring, on page 4](#)
- [Restrictions for traffic mirroring, on page 4](#)
- [Traffic mirroring terminology, on page 5](#)
- [Traffic mirroring types, on page 6](#)
- [How traffic mirroring works, on page 6](#)

Traffic mirroring

Traffic mirroring or port mirroring is a network monitoring feature that

- monitors Layer 3 network traffic passing in or out of Ethernet interfaces
- directs this traffic to a network analyzer for analysis, and
- operates as a proprietary function under Cisco Switched Port Analyzer (SPAN).

Table 2: Feature History Table

Feature Name	Release Information	Feature Description
Traffic Mirroring	Release 25.1.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100]) Traffic mirroring, also known as port mirroring or Switched Port Analyzer (SPAN), is a proprietary feature by Cisco. It allows the monitoring of Layer 3 network traffic that enters or exits a set of Ethernet interfaces. This mirrored traffic can then be directed to a network analyzer for further analysis. This feature is now supported on: • 8011-4G24Y4H-I

Benefits of traffic mirroring

Traffic mirroring allows you to

- monitor Layer 3 network traffic
- analyze traffic using a network analyzer, and
- mirror traffic without affecting the switching of traffic on source interfaces.

Configuration guidelines for traffic mirroring

These guidelines apply to traffic mirroring:

- Supports mirroring traffic on a GRE IPv4 or IPv6 tunnel, also known as Encapsulated Remote Switched Port Analyzer (ERSPAN).
- The system allows eight monitor sessions for ERSPAN, four monitor sessions for local SPAN, and four monitor sessions for SPAN to file.
- The total number of monitor sessions for all SPAN features is eight.

Restrictions for traffic mirroring

These restrictions apply to traffic mirroring:

- The router does not support traffic mirroring counters per interface.
- The router does not support bundle member interfaces as sources for mirroring sessions.

- The router does not support port-level mirroring for any type of SPAN.
- ERSPAN tunnel statistics is not supported.
- The dropped packets at NPU cannot be captured by regular ERSPAN session. For capturing dropped packets at NPU, use the mirror forward-drop packet feature.
- From Release 24.3.1, layer 2 SPAN features are supported on PWHE. For more information about the layer 2 SPAN features supported on PWHE, see the *Traffic Mirroring on PWHE* section in the *L2VPN Configuration Guide for Cisco 8000 Series Routers*.

Traffic mirroring terminology

Understanding traffic mirroring terminology is essential for effective configuration and troubleshooting.

These are the key terms associated with traffic mirroring:

- Ingress traffic: Traffic that enters the router.
- Egress traffic: Traffic that leaves the router.
- Source port: A port that the system monitors using traffic mirroring. It is also known as monitored port.
- Destination port: A port that monitors the source ports. Usually, a network analyzer is connected here.
- Monitor session: A collection of traffic mirroring configurations that consists of a single destination and many source interfaces.

Source port

A source port, as used in traffic mirroring, is a switched or routed port that

- is monitored for network traffic analysis
- supports monitoring of ingress or Rx traffic in a single local or remote mirroring session, and
- supports source ports up to 800.

Characteristics of source port

A source port has these characteristics:

- It supports any port type, such as Bundle Interface, sub-interface, 100-Gigabit Ethernet, and 400-Gigabit Ethernet.
- It does not support Bridge group virtual interfaces (BVI).
- It can only be monitored in one traffic mirroring session at a time.
- It cannot be a destination port.
- It can be configured with a direction, ingress for monitoring. The direction applies to all physical ports in a bundle.

Monitor session

In traffic mirroring, a monitor session is a collection of traffic mirroring configurations that

- consists of a single destination and can have several source interfaces
- sends traffic from source interfaces or ports to a monitoring or a destination port
- combines mirrored traffic streams at the destination port when multiple source ports are present, and
- results in a mixture of traffic from one or more source ports at the destination port.

Characteristics of monitor session

Monitor sessions have these characteristics:

- Each monitor session can have only one destination port.
- Each destination port can belong to only one monitor session.
- The destination of an ERSPAN monitoring session is either a GRE IPv4 or a GRE IPv6 tunnel.

Traffic mirroring types

The router supports these traffic mirroring types:

- [ACL-based traffic mirroring](#)
- [Encapsulated Remote Switched Port Analyzer \(ERSPAN\)](#)
- [Local SPAN](#)
- [SPAN to file](#)
- [Mirror forward-drop packets](#)
- [Mirror buffer-drop packets](#)
- [File mirroring](#)

**Note**

Starting with Cisco IOS XR Software Release 7.0.14, the configurations of ERSPAN and security ACLs do not impact each other and have no dependencies. Both can be applied simultaneously.

How traffic mirroring works

Traffic mirroring consists of these key operations:

- copy traffic from Layer 3 interfaces or sub-interfaces.
- send copied traffic to a destination for analysis.

- use traffic mirroring ports configured to receive packet copies.

Traffic mirroring functions differently on switches and hubs due to a fundamental difference.

Summary

These are the key components involved in traffic mirroring:

- Copying traffic: Duplicating network traffic from specified sources.
- Source of traffic: Traffic is copied from Layer 3 interfaces or sub-interfaces.
- Destination for analysis: The copied traffic is sent to a designated location . For example, a traffic analyzer for evaluation.
- Traffic mirroring ports: Specific ports configured to receive these packet copies.
- Unicast traffic capture: The process enables the capture of unicast traffic, which is particularly important for devices like switches and routers that typically direct unicast packets to specific ports.

Traffic mirroring copies network traffic from Layer 3 interfaces to designated ports for analysis, enabling the capture of unicast traffic. On hubs, all ports receive packet copies, allowing easy traffic capture. In contrast, switches and routers require traffic mirroring to analyze unicast traffic since they direct packets to specific ports based on MAC address learning. Configuring mirroring ports allows analyzers to evaluate network traffic effectively.

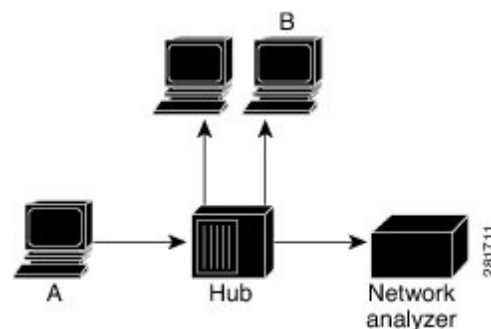
These stages describe traffic mirroring:

- MAC address learning - the switch or router learns the MAC address of Host B.
- Forwarding decision - the switch or router decides to forward unicast traffic from Host A to Host B based on the learned MAC address.
- Traffic forwarding - the unicast traffic is forwarded to the specific port associated with Host B.
- Traffic analyzer visibility - this unicast traffic escapes detection by the traffic analyzer because the traffic directs itself to a specific port and does not broadcast.

Consider this example, where, if you want to capture Ethernet traffic that is sent by host A to host B, and both are connected to a hub, attach a traffic analyzer to the hub. All other ports see the traffic between hosts A and B.

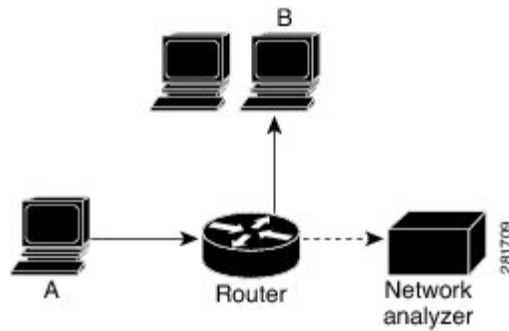
Workflow

Figure 1: Traffic mirroring operation on a hub



Consider this example, where, the traffic analyzer captures only traffic that is flooded to all ports.

Figure 2: Network analysis does not work on a router without traffic mirroring



In this configuration, the traffic analyzer captures only traffic that is flooded to all ports, including:

- broadcast traffic
- multicast traffic with CGMP or Internet Group Management Protocol (IGMP) snooping disabled, and
- unknown unicast traffic on a switch.

An extra feature is necessary that artificially copies unicast packets that host A sends. This extra feature is traffic mirroring. When traffic mirroring is enabled, the traffic analyzer is attached to a port that is configured to receive a copy of every packet that host A sends. This port is called a traffic mirroring port. The other sections of this document describe how you can fine tune the traffic mirroring feature.

When...	Then...
a hub receives a packet on one port	the hub sends out a copy of that packet from all ports except the one where the packet was received.
a switch boots up	the switch begins building a Layer 2 forwarding table based on the source MAC addresses of received packets.
the switch builds its Layer 2 forwarding table	the switch forwards traffic destined for a MAC address directly to the corresponding port.



CHAPTER 3

ACL-based traffic mirroring

This chapter introduces ACL-based traffic mirroring, a feature for selectively monitoring network traffic by using ACL configurations. It covers benefits like improved security and efficient resource use, details configuration guidelines and restrictions, and provides step-by-step procedures for setting up IPv4, IPv6, and Layer 3 ACLs.

- [ACL-based traffic mirroring, on page 9](#)
- [Configuration guidelines for ACL-based traffic mirroring, on page 10](#)
- [Restrictions for ACL-based traffic mirroring, on page 10](#)
- [How ACL-based traffic mirroring works, on page 10](#)
- [Configure ACLs for traffic mirroring, on page 11](#)
- [Configure layer 3 ACL-based traffic mirroring, on page 12](#)
- [Attach a source interface, on page 12](#)
- [Multiple SPAN ACL sessions for MPLS, on page 14](#)

ACL-based traffic mirroring

ACL-based traffic mirroring is a feature that allows you to:

- mirror traffic based on the configuration of the global interface ACL, and
- monitor particular types of traffic that match certain characteristics, such as source or destination IP addresses, protocols, or port numbers.

Benefits of ACL-based traffic mirroring

- **Selective monitoring:** Allows you to focus on specific traffic flows that are of interest, thereby reducing the amount of data that needs to be processed and analyzed.
- **Improved security:** Enables the monitoring of suspicious or critical traffic patterns, helping to detect and respond to potential security threats more effectively.
- **Efficient resource usage:** Mirrors only selected traffic and uses network resources such as bandwidth and processing power more efficiently, avoiding the overhead of capturing all traffic.

Configuration guidelines for ACL-based traffic mirroring

These configuration guidelines apply to ACL-based traffic mirroring:

Traffic mirroring on source interface

- Configure ACLs on the source interface to avoid default mirroring of traffic.
- Configure the ACLs on the bundle interface and not on bundle members, if a bundle interface is a source interface.

Restrictions for ACL-based traffic mirroring

These restrictions apply to ACL-based traffic mirroring:

- You must remove and re-apply monitor-sessions on all interfaces after modifying the access control list (ACL).
- SPAN ACL does not support User Defined Fields (UDF).
- Deny action in SPAN ACL is ignored, and no packet drops from SPAN ACL. Deny ACEs will be internally converted to permit ACEs. Packets will also be mirrored.
- There is no implicit deny-all entry in SPAN ACL.
- IPV6 ACL is required for mirroring IPv6 packets and IPv4 ACL is required for IPv4 packets. This follows the same structure as security ACL with IPv4 and IPv6 mirror options.

How ACL-based traffic mirroring works

These stages describe how ACL-based traffic mirroring works.

1. The router creates an ACL.
2. This ACL defines the traffic to be mirrored and specifies the packets to be mirrored.
3. The keywords defined in the ACL designates that the packet is mirrored to the destination port.
4. The rules defined in the ACL determine the behavior for traffic.

When the system..	But..	Then..
configures the acl command on the source mirroring port	the ACL configuration command does not use the capture keyword	the router does not mirror the traffic.
uses ACL configuration with the capture keyword	you have not configured the acl command on the source port	the router mirrors the traffic, but does not apply access list configuration.

Configure ACLs for traffic mirroring

Use this procedure to configure IPv4 or IPv6 ACLs for traffic mirroring.

Procedure

Step 1 Create SPAN IPv4 ACL for traffic mirroring.

Example:

```
Router(config)# ipv4 access-list v4-monitor-acl
Router(config-ipv4-acl)# 10 permit udp 20.1.1.0 0.0.0.255 eq 10 any
Router(config-ipv4-acl)# 20 permit udp 30.1.1.0 0.0.0.255 eq 20 any
Router(config-ipv4-acl)# exit
Router(config)# commit
```

The router creates an ACL named **v4-monitor-acl** and applies the **permit** action for the traffic.

Note

If you specify **deny** action, the router drops the traffic for that interface. Mirroring happens only if you add the **icmp-off** keyword to the ACE as shown. You can use the **icmp-off** keyword only for security or hybrid ACLs.

```
ipv4 access-list acl1
10 deny ipv4 any 2.1.0.0/16 capture icmp-off
20 permit ipv4 any any
!
```

Step 2 Create SPAN IPv6 ACL for traffic mirroring.

Example:

```
Router(config)# ipv6 access-list v6-monitor-acl
Router(config-ipv6-acl)# 10 permit ipv6 host 120:1:1::1 host 130:1:1::1
Router(config-ipv6-acl)# exit
```

The router creates an ACL named **v6-monitor-acl** and applies the **permit** action for the traffic.

Step 3 Apply the traffic monitoring to SPAN source interface.

Example:

```
Router(config)# interface HundredGigE0/0/0/12
Router(config-if)# monitor-session mon1 ethernet direction rx-only
Router(config-if)# acl ipv4 v4-monitor-acl
Router(config-if)# acl ipv6 v6-monitor-acl!
!
```

For **v4-monitor-acl** and **v6-monitor-acl** ACLs, the router applies traffic mirroring for **HundredGigE0/0/0/12** interface.

Note

To enable traffic mirroring, include the **capture** keyword for security or hybrid ACLs.

Step 4 Verify the ACL configuration on your router.

Example:

```
Router# show access-lists ipv4 v4span1 hardware ingress span interface HundredGigE0/0/0/12
location 0/3/cpu0
```

```
ipv4 access-list v4-monitor-acl
10 permit ipv4 host 51.0.0.0 host 101.0.0.0
20 permit ipv4 host 51.0.0.1 host 101.0.0.1
30 permit ipv4 host 51.0.0.2 any
40 permit ipv4 any host 101.0.0.3
50 permit ipv4 51.0.1.0 0.0.0.255 101.0.1.0 0.0.0.255
60 permit ipv4 51.0.2.0 0.0.0.255 101.0.2.0 0.0.0.255 precedence critical
```

Configure layer 3 ACL-based traffic mirroring

Use these steps to configure traffic mirroring for layer 3 ACLs.

Procedure

Step 1 Start a monitor session.

Example:

```
Router# configure
Router(config)# monitor-session ms1
```

Step 2 Define a destination.

Example:

```
Router(config-mon)# destination tunnel-ip 1
Router(config-mon)# commit
```

Step 3 Attach an interface to the monitor session for an acl and specify the direction for which you want to mirror the traffic.

Example:

```
Router# configure
Router(config)# interface HundredGigE/2/0/11
Router(config-if)# ipv4 access-group span ingress
Router(config-if)# monitor-session ms1 ethernet direction rx-only acl
Router(config-if-mon)# commit
```

Step 4 Use the **capture** keyword to start mirroring traffic for the ACL.

Example:

```
Router# configure
Router(config)# ipv4 access-list span
Router(config-ipv4-acl)# 5 permit ipv4 any any dscp 5 capture
Router(config-ipv4-acl)# 10 permit ipv4 any any
Router(config-ipv4-acl)# commit
```

Attach a source interface

Use these steps to attach a source interface to a monitor session for an ACL.

Procedure

Step 1 Enter global configuration mode.

Example:

```
Router# configure
```

Step 2 Enter interface configuration mode for the specified source interface in **interface** *type number*. The interface number is entered in *rack/slot/module/port* notation. For more information about the syntax for the router, use the question mark (?) online help function.

Example:

```
Router(config)# interface HundredGigE 0/1/0/10/0/1/0
```

Step 3 Control access to an interface in **ipv4 access-group** *acl-name* {**ingress** | **egress**}.

Example:

```
Router(config-if)# ipv4 access-group acl1 ingress
```

Step 4 Attach a monitor session to the source interface and enter the monitor session configuration mode.

Example:

```
Router(config-if)# monitor-session mon1 ethernet direction rx-only port-level acl
Router(config-if-mon)#
```

Note

rx-only specifies that only ingress traffic is replicated.

Step 5 Specify that the traffic mirrored is according to the defined ACL.

Example:

```
Router(config-if-mon)# acl
```

Note

If an ACL is configured by name, it overrides any ACL that may be configured on the interface.

Step 6 Exit the monitor session configuration mode and return to interface configuration mode.

Example:

```
Router(config-if-mon)# exit
Router(config-if)#
```

Step 7 Save configuration changes.

Example:

```
RP/0/RP0/CPU0(config-if)# end
or
RP/0/RP0/CPU0(config-if)# commit
```

When you execute the **end** command, the system prompts you to commit changes.

```
Uncommitted changes found, commit them before exiting (yes/no/cancel)?
[cancel]:
```

- **yes** - Entering **yes** saves configuration changes to the running configuration file, exits the configuration session, and returns the router to EXEC mode.

- **no** - Entering **no** exits the configuration session and returns the router to EXEC mode without committing the configuration changes.
- **cancel** - Entering **cancel** leaves the router in the current configuration session without exiting or committing the configuration changes.
- **commit** - Use the **commit** command to save the configuration changes to the running configuration file and remain within the configuration session.

Step 8 Display information about the monitor session.

Example:

```
Router# show monitor-session status
```

Multiple SPAN ACL sessions for MPLS

Multiple SPAN ACL sessions for MPLS is a feature that

- monitors the MPLS traffic by configuring multiple SPAN ACL sessions for MPLS, and
- mirrors ingress MPLS traffic through the **monitor-session session-name ethernet direction rx-only port-level** command.

Table 3: Feature History Table

Feature Name	Release Information	Description
Multiple SPAN ACL sessions for MPLS	Release 25.1.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100])(select variants only*) *This feature is supported on Cisco 8011-4G24Y4H-I routers.

Feature Name	Release Information	Description
Multiple SPAN ACL sessions for MPLS	Release 24.4.1	Introduced in this release on: Fixed Systems(8200, 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*). This feature verifies the overall network performance simultaneously from various network locations and ensures a better network visibility, network resource efficiency, and flexibility. This feature allows to configure multiple SPAN ACL sessions for MPLS on Layer 3 interfaces configured on the Label-Switched Paths (LSPs) to monitor the MPLS traffic based on the labels and the EXP bit. This MPLS SPAN ACL configuration is supported only in the ingress direction. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 8712-MOD-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH This feature introduces these changes: CLI: • acl mpls • mpls access-list YANG Data Model: Cisco-IOS-XR-um-mpls-acl-cfg.yang (see Github, YANG Data Models Navigator).

Starting from Cisco IOS XR Release 24.4.1, you can monitor the MPLS traffic by configuring multiple SPAN ACL sessions for MPLS. With this feature, the ingressing MPLS traffic is mirrored. This is achieved with the **monitor-session session-name ethernet direction rx-only port-level** configuration.

This feature is supported on both the Physical and Bundle main and subinterfaces.

Benefits of using multiple MPLS SPAN ACL sessions

We recommend using multiple SPAN ACL sessions for MPLS because this approach:

- Improves flexibility of the associated user interface.

- Avoids redundancy.
- Provides backward compatibility.
- Minimises configuration size on the disk.
- Reduces process memory in both the shared plane and local plane for scale configurations.

Configuration guidelines for multiple MPLS SPAN ACL sessions

Follow these guidelines when configuring multiple SPAN ACL sessions for MPLS:

- Configure multiple ACL sessions for MPLS exclusively in the ingress (Rx) direction.
- Configure up to four SPAN sessions.
- Do not use the *Deny* action as it is not supported.
- Use only GRE tunnel interfaces as the destination interfaces.
- Specify the monitor sessions for the configured interfaces.
- Use the SPAN session ID to distinguish between multiple SPAN sessions under the same source interface.

Configure multiple SPAN ACL sessions for MPLS

Use these steps to configure multiple SPAN ACL sessions for MPLS:

Procedure

- Step 1** Define multiple SPAN ACLs for the incoming (Rx) traffic or the MPLS packets captured. In this example, multiple SPAN ACLs, mp1 and mp2, are defined for mirroring MPLS traffic.

Example:

```
Router(config)# mpls access-list mp1
Router(config-mpls-acl)# 10 permit label1 2000 label2 3000 label3 4000 exp1 5 exp2 5
exp3 7
Router(config-mpls-acl)# exit
Router(config)# mpls access-list mp2
Router(config-mpls-acl)# 10 permit label3 9000 exp3 5
Router(config-mpls-acl)# exit
Router(config)# commit
```

- Step 2** Configure a monitor session on the specified destination interface for the incoming (Rx) traffic.

Example:

```
Router#config
Router(config)#interface tunnel-ip41
Router(config-if)#tunnel source 11.11.11.11
Router(config-if)#tunnel destination 22.22.22.22
Router(config-if)#ipv4 address 41.41.41.2 255.255.255.0
Router(config-if)#tunnel mode gre ipv4
Router(config-if)#commit
Router(config-if)#exit
!
```

```
Router(config)#monitor-session S1 ethernet destination interface tunnel-ipv41
Router(config-if)#commit
!
```

- Step 3** Attach monitor session to source interface. This configuration attaches the MPLS SPAN ACL sessions to the specified source interface. Use the **direction rx-only** keyword so that only the ingress traffic is mirrored.

Example:

```
Router(config)# interface tenGigE 0/0/0/14
Router(config-if)# monitor-session S1 ethernet direction rx-only port-level
Router(config-if-mon)# acl mpls mp1
!!
```

- Step 4** Execute this command to view the running configuration.

Example:

```
Router# show running-config interface tenGigE 0/0/0/14
Mon Apr 1 13:16:47.430 UTC
interface TenGigE0/0/0/14
  ipv4 address 1.1.1.1 255.255.255.0
  ipv6 address 1111::1:1/96
  monitor-session S1 ethernet direction rx-only port-level
  acl mpls mp1
!
Router#
```

- Step 5** Verify the monitor session and details of the session.

Example:

```
Router# show monitor-session status
Mon Apr 1 13:16:40.408 UTC
Monitor-session S1
Destination interface tunnel-ip41
=====
Source Interface Dir Status
-----
Te0/0/0/14 (port) Rx Operational

Router# show monitor-session status detail
Mon Apr 1 13:19:11.124 UTC
Monitor-session S1
Destination interface tunnel-ip41
Source Interfaces
-----
TenGigE0/0/0/14
Direction: Rx-only
Port level: True
ACL match: Disabled
IPv4 ACL: Disabled
IPv6 ACL: Disabled
MPLS ACL: Enabled (mp1)
Portion: Full packet
Interval: Mirror all packets
Mirror drops: Disabled
Status: Operational
RP/0/RP0/CPU0:ios#
```




CHAPTER 4

Local SPAN

This chapter details a traffic mirroring feature that allows users to monitor network traffic on a single router or switch. It explains how Local SPAN copies traffic from source interfaces or VLANs to a destination port on the same device for analysis.

- [Local SPAN, on page 19](#)
- [Configuration guidelines for local SPAN, on page 20](#)
- [Restrictions for local SPAN, on page 21](#)
- [Configure local SPAN, on page 21](#)
- [Local SPAN with ACLs, on page 23](#)
- [Local SPAN rate limit, on page 25](#)

Local SPAN

Local SPAN is a traffic mirroring feature and the most basic form of traffic mirroring that

- mirrors both source and destination interfaces on the same router, and
- simplifies the router setup and reduces the need for additional hardware.

Table 4: Feature History Table

Feature Name	Release Information	Description
Local SPAN	Release 25.4.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100])(select variants only*) *This feature is supported on: • 8011-32Y8L2H2FH • 8011-12G12X4Y-A • 8011-12G12X4Y-D

Feature Name	Release Information	Description
Local SPAN	Release 25.2.1	Introduced in this release on: Modular Systems (8800 [LC ASIC: P100])(select variants only*) *This feature is now supported on: • 88-LC1-36EH • 88-LC1-12TH24FH-E
Local SPAN	Release 25.1.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100]) Local SPAN allows you to monitor network traffic on a single switch by copying traffic from one or more source ports or VLANs to a destination port for analysis. This process occurs entirely within the same device, ensuring that both source and destination ports are located on it. You can monitor and analyze traffic without affecting the normal operation of the network. This feature is now supported on: • 8011-4G24Y4H-I

Configuration guidelines for local SPAN

The local SPAN feature supports multiple capabilities. These are the key guidelines:

- You can only configure this feature for ingress traffic.
- Configures the destination interface only as Layer 2 or Layer 3 physical main interface.
- Configures these interfaces as sources for a local SPAN session:
 - L3 physical main and sub-interface, bundle main, and bundle sub-interface.
 - L2 ethernet interfaces: Ethernet Flow Point (EFP) and trunk.
 - BVI interface.
- Mirrors IPv4, IPv6, IP-in-IP and MPLS traffic.
- Supports extended Access Control Lists (ACLs) to reduce mirrored traffic throughput.
- Supports traffic shaping on the destination interface.
- Supports session statistics. There is one counter for all types of traffic, that is, IPv4, IPv6, and MPLS.
- Supports up to four local SPAN sessions, shared between:
 - ERSPAN
 - Local SPAN, and
 - SPAN to File features.

- Supports up to 1000 source interfaces.

Restrictions for local SPAN

Generic restrictions for local SPAN

The generic restrictions for local SPAN include:

- Egress mirroring isn't supported.
- The physical interface used as destination can't be a bundle member link.
- GRE tunnels are not supported as source or destination interfaces.
- Per-source interface mirroring statistics isn't supported. However, SPAN session statistics are supported. The session statistics contain the total number of packets mirrored by the session.
- A destination interface can't be a mirrored source interface and vice versa.
- NetFlow or sFlow configuration is not supported on interfaces that already have a local SPAN session configured.
- The dropped packets at NPU cannot be captured by regular SPAN session. For capturing dropped packets at NPU, use the mirror forward-drop packets feature.

ACL restrictions for local SPAN

The ACL restrictions for local SPAN include:

- ACLs for Local SPAN are applied only in ingress direction.
- If the ACL keyword is present in monitor-session configuration for an interface but no ACL is applied to that interface, traffic packets are not mirrored.
- ACL for MPLS traffic isn't supported.

Configure local SPAN

Use these steps to configure local SPAN.

Procedure

Step 1 Create a local SPAN session.

Example:

```
Router#configure
Router(config)#monitor-session mon1 ethernet
Router(config-mon)#destination interface HundredGigE0/1/0/0
Router(config-mon)#commit
```

```
Router(config-mon)#end
Router#
```

Step 2 Attach the SPAN session to an interface.

Example:

```
Router(config-mon)#interface HundredGigE0/1/0/2
Router(config-if)# monitor-session mon1 ethernet direction rx-only
Router(config-if-mon)# no shut
Router(config-if)#!
Router(config-if)#
Router(config-if)#interface Bundle-Ether1
Router(config-if)# monitor-session mon1 ethernet direction rx-only
Router(config-if-mon)# no shutdown
Router(config-if)#!
Router(config-if)#

Router:monitor(config-if)#interface HundredGigE0/1/0/14.100
Router:monitor(config-subif)# monitor-session mon1 ethernet direction rx-only
Router:monitor(config-if-mon)# no shut
Router:monitor(config-subif)#!
Router:monitor(config-subif)#
Router:monitor(config-subif)#interface Bundle-Ether1.1
Router:monitor(config-subif)# monitor-session mon1 ethernet direction rx-only
Router:monitor(config-if-mon)# no shut
Router:monitor(config-subif)#!
Router:monitor(config-subif)#commit
```

Step 3 Verify the configuration.

Example:

```
Router#show monitor-session status
Monitor-session mon1
Destination interface HundredGigE0/1/0/0
=====
Source Interface Dir Status
-----
Hu0/1/0/2 Rx Operational
Hu0/1/0/14.100 Rx Operational
BE1 Rx Operational
BE1.1 Rx Operational
```

Step 4 Execute the **show monitor-session status internal** command for session statistics.

Example:

```
Router#show monitor-session status internal
Thu Aug 13 20:05:23.478 UTC
Information from SPAN Manager and MA on all nodes:
Monitor-session mon1 (ID 0x00000001) (Ethernet)
SPAN Mgr: Destination interface HundredGigE0/1/0/0 (0x00800190)
Last error: Success
0/1/CPU0: Destination interface HundredGigE0/1/0/0 (0x00800190)
0/RP0/CPU0: Destination interface HundredGigE0/1/0/0 (0x00800190)
Information from SPAN EA on all nodes:
Monitor-session 0x00000001 (Ethernet)
0/1/CPU0: Name 'mon1', destination interface HundredGigE0/1/0/0 (0x00800190)
Platform, 0/1/CPU0:
Monitor Session ID: 1
Monitor Session Packets: 32
Monitor Session Bytes: 4024
0/2/CPU0: Name 'mon1', destination interface HundredGigE0/1/0/0 (0x00800190)
Platform, 0/2/CPU0:
```



```
Monitor Session ID: 1
Monitor Session Packets: 0
Monitor Session Bytes: 0
```

Local SPAN with ACLs

Local SPAN with an Access Control List (ACL) is a traffic mirroring feature that:

- filters and mirrors ingress traffic
- considers only Access Control Entries (ACEs) with capture keyword for traffic mirroring
- captures both permit and deny packets if the ACE contains the **capture** keyword, and
- allows one IPv4 ingress ACL and one IPv6 ingress ACL per interface.

Benefits of local SPAN with ACL

These are the benefits of local SPAN with ACL:

- **Traffic Filtering:** Local SPAN with ACL allows precise filtering of ingress traffic, ensuring that only relevant data is mirrored based on specific criteria.
- **Selective Mirroring:** By using the **capture** keyword in ACEs, you can selectively mirror both permitted and denied packets, providing flexibility in monitoring.
- **Efficient Monitoring:** Specifying one IPv4 and one IPv6 ingress ACL per interface streamlines monitoring processes and reduces unnecessary data capture.
- **Enhanced Network Security:** Filtering and mirroring ingress traffic can help identify and analyze potential security threats or anomalies in network traffic.

Configure local SPAN with IPv4 ACLs

Use these steps to configure local SPAN using IPv4 ACLs.

Procedure

Step 1 Configure IPv4 ACLs for traffic mirroring.

Example:

```
Router(config)# ipv4 access-list acl1
Router(config-ipv4-acl)# 10 permit ipv4 25.0.0.0 0.0.0.255 any capture
Router(config-ipv4-acl)# 20 permit ipv4 20.0.0.0 0.0.0.255 any
Router(config-ipv4-acl)# 30 permit ipv4 131.1.1.0 0.0.0.255 any capture
Router(config-ipv4-acl)# 40 permit ipv4 191.1.1.0 0.0.0.255 any capture
```

Step 2 Apply the traffic monitoring to an interface.

Example:

```
Router(config)# interface HundredGigE0/1/0/2
Router(config-if)# ipv4 address 131.1.1.2 255.255.255.0
Router(config-if)# monitor-session mon1 ethernet direction rx-only port-level
Router(config-if-mon)# acl
Router(config-if-mon)# ipv4 access-group acl1 ingress
```

Step 3 Verify the configuration.

Example:

```
Router#show running-config ipv4 access-list acl1
Thu Aug 13 20:22:54.388 UTC
ipv4 access-list acl1
10 permit ipv4 22.0.0.0 0.0.0.255 any capture
20 permit ipv4 20.0.0.0 0.0.0.255 any
30 permit ipv4 131.1.1.0 0.0.0.255 any capture
40 deny ipv4 181.1.1.0 0.0.0.255 any capture
!
```

Configure local SPAN with IPv6 ACLs

Use these steps to configure local SPAN using IPv6 ACLs.

Procedure

Step 1 Configure IPv6 ACLs for traffic mirroring.

Example:

```
Router(config)# ipv6 access-list acl2
Router(config-ipv6-acl)# 10 permit ipv6 10:1:1::2/64 any capture
Router(config-ipv6-acl)# 20 permit ipv6 10:1:1::3/64 any
Router(config-ipv6-acl)# 30 permit ipv6 10:1:1::4/64 any capture
```

Step 2 Apply the traffic monitoring to an interface.

Example:

```
Router(config)# interface HundredGigE0/1/0/3
Router(config-if)# ipv6 address 10:1:1::5/64
Router(config-if)# monitor-session mon2 ethernet direction rx-only port-level
Router(config-if-mon)# acl
Router(config-if-mon)# ipv6 access-group acl2 ingress
```

Step 3 Verify the configuration.

Example:

```
Router#show running-config ipv6 access-list acl2
Thu Aug 14 20:22:54.388 UTC
ipv6 access-list acl2
10 permit ipv6 10:1:1::2/64 any capture
20 permit ipv6 10:1:1::3/64 any
30 permit ipv6 10:1:1::4/64 any capture
!
```

Local SPAN rate limit

Local SPAN rate limit is a traffic mirroring feature that

- takes place at the session level, not at source interface level
- configures rate limiting when you configure a traffic class in local SPAN session
- uses traffic class to shape traffic on an egress interface, and
- applies a Quality of Service (QoS) policy to the egress interface over which the router send the mirrored traffic.

Example

This example provides details on how to configure the local SPAN rate limit. In this example, the class, TC5, matches with the class that you configured in the monitor session.

```
Router# monitor-session mon2 ethernet
destination interface HundredGigE0/1/0/19
traffic-class 5
class-map match-any TC5
match traffic-class 5
end-class-map
policy-map shape-foo
class TC5 /* This has to match the class that was configured on monitor session */
shape average percent 15
class class-default
interface HundredGigE0/1/0/19 /* This is the egress interface over which mirrored packets
are sent */
service-policy output shape-foo
```




CHAPTER 5

ERSPAN

This chapter, Encapsulated Remote Switched Port Analyzer (ERSPAN), explains its function in mirroring network traffic via GRE tunnels for analysis. It covers features like higher payload support, GRE IPv6 and MPLS integration, partial packet capture, rate limiting, DSCP-based traffic classification, and multi-session monitoring with ACLs.

- [ERSPAN, on page 27](#)
- [Configuration guidelines for ERSPAN, on page 30](#)
- [Restrictions for ERSPAN, on page 31](#)
- [Supported ERSPAN sessions, on page 32](#)
- [ERSPAN over GRE, on page 33](#)
- [Partial packet capture for ERSPAN, on page 36](#)
- [ERSPAN with flexible CLI, on page 38](#)
- [ERSPAN rate limit, on page 38](#)
- [ERSPAN rate-limit per destination , on page 42](#)
- [Traffic mirroring with DSCP, on page 44](#)
- [Monitor ERSPAN sessions with SPAN and security ACLs, on page 51](#)
- [Selective packet capture for protocol and drops, on page 53](#)

ERSPAN

Encapsulated Remote Switched Port Analyzer (ERSPAN) is a traffic mirroring feature that

- monitors network traffic from one or more source ports on a router
- mirrors traffic from one or more source ports, and
- sends the mirrored traffic through GRE tunnels to destinations for analysis.

The destination may be a network analyzer or other monitoring devices.

Table 5: Feature History Table

Feature Name	Release Information	Feature Description
Higher payload analysis with eight ERSPAN sessions	Release 25.4.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100])(select variants only*) *This feature is now supported on: • 8011-12G12X4Y-A • 8011-12G12X4Y-D
Higher payload analysis with eight ERSPAN sessions	Release 25.1.1	Introduced in this release on: Fixed Systems (8010 [ASIC: A100]) This feature is now supported on Cisco 8011-4G24Y4H-I routers.
ERSPAN over GRE IPv6	Release 25.1.1	Introduced in this release on: Fixed Systems (8700 [ASIC: K100] , 8010 [ASIC: A100]) This feature is now supported on: • 8712-MOD-M • 8011-4G24Y4H-I
Higher payload analysis with eight ERSPAN sessions	Release 24.4.1	Introduced in this release on: Fixed Systems(8200 , 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100]) (select variants only*). This feature now enables the Cisco 8000 Series routers to support eight ERSPAN sessions on the following hardware thus allowing you to analyze higher payloads in real time across Layer 3 domains on your network. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 8712-MOD-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH

Feature Name	Release Information	Feature Description
ERSPAN over GRE IPv6	Release 24.4.1	Introduced in this release on: Fixed Systems(8200 , 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100]) (select variants only*). With this release, the router allows you to mirror IPv4 or IPv6 traffic with ERSPAN over GRE IPv6 sessions to monitor traffic on remote traffic analyzers on the following hardware. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 8712-MOD-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH
Partial packet capture ability for ERSPAN (Rx)	Release 7.5.3	With this feature, you can perform partial packet capture in the Rx direction. Earlier, the ability for entire packet capture was available, now you can choose entire or partial packet capture in the Rx direction. Here, partial packet capture is also known as truncation.
ERSPAN over MPLS traffic	Release 7.3.5 Release 7.5.3	With this release, the router allows you to mirror MPLS traffic and set up the GRE tunnel with the next hop over a labeled path. This feature helps you to remote-monitor the traffic on traffic analyzers.

Benefits of ERSPAN

These are the benefits of ERSPAN:

- It enables real-time network troubleshooting.
- It automates configuration processes, allowing network operators to troubleshoot issues in real-time and send specific flows to management servers for detailed analysis.

- It transports mirrored traffic over an IP network and encapsulates it at the source router for transfer across the network

Configuration guidelines for ERSPAN

These guidelines apply for ERSPAN:

Source and destination interface requirements

- The source interfaces are Layer 3 interfaces, such as physical and bundle interfaces, or subinterface.
- Each monitor session allows only one destination interface.
- The next hop interface must be a main interface. It can be a physical or bundle interface.

Traffic types and protocol support

- The routers mirror IPv4 and IPv6 traffic.
- The router supports MPLS traffic mirroring and GRE tunnel configuration with the next hop over a labeled path.
- The router supports only ERSPAN TYPE II header. The value of the index field is always 0. The value of the session-ID field is an internal number that is used by the data path to distinguish between sessions.
- The router mirrors only unicast traffic. However, from Cisco IOS XR Software Release 7.5.3 onwards, the router can mirror multicast traffic.
- ERSPAN will be functional regardless of any configuration related to MPLS or LDP present on the router.

Encapsulation and tunneling

- ERSPAN with GRE IPv4 or IPv6 has tunnel destinations.
- In ERSPAN over GRE IPv6, the **HopLimit** and **TrafficClass** fields in outer IPv6 header can be edited under the tunnel configuration.

Mirroring support

- ERSPAN supports only Rx direction.
- MPLS packet mirroring is supported only from Cisco IOS XR Software Release 7.5.3 onwards.

Access Control List (ACL) integration

- ERSPAN over GRE IPv4 and IPv6 supports SPAN ACL.
- For ACL ERSPAN, the ERSPAN next-hop must have ARP resolved. Any other traffic or protocol triggers ARP.
- ACL permit or deny entries with capture action are part of mirroring features.

Packet capture capabilities

The router supports full packet capture.

Monitor session support

- Starting from Cisco IOS XR Release 24.3.1, the system creates one default monitor session and users can configure up to three additional monitor sessions, totaling four sessions, which is the maximum number of monitor sessions that the router allows.

Release-wise feature summary

This table summarizes key feature changes across different software releases.

Table 6: Release-wise feature summary

Feature	Release	Details
GRE sequence bit for ERSPAN	7.0.14	GRE header sequence bit is set; sequence number is always 0 for ERSPAN packets.
Sequence bit	7.5.3	The sequence number bit is always set one and the sequence number field, which is 4 bytes, is always set to zero.
Packet truncation over remote GRE tunnels	7.5.3	Allows flexible packet truncation for mirrored traffic.

Restrictions for ERSPAN

These restrictions apply for ERSPAN

- GRE tunnels are exclusively dedicated to ERSPAN mirrored packets. Do not configure any IPv4 and IPv6 addresses under the GRE tunnel.
- Traffic accounting of the ERSPAN mirrored packets is not supported.



Note You can view the SPAN packet count per session, using the [show monitor-session status internal](#) command.

- ERSPAN decapsulation is not supported.
- ERSPAN does not work if the GRE next hop is reachable over subinterface. For ERSPAN to work, the next hop must be reachable over the main interface. From Cisco IOS XR Software Release 7.5.3 onwards, GRE next hop can be resolved over subinterface or the main interface.
- ERSPAN will be functional regardless of any configuration related to MPLS or LDP present on the router.
- MPLS packet mirroring is supported only from Cisco IOS XR Software Release 7.5.3 onwards.

- On Cisco Silicon One Q100 ASIC-based systems, only the upper 64-bits of the source IPv6 address in the outer IPv6 header of an ERSPAN packet are valid because of data path limitations; the lower 64-bits are set to zero. The destination GREv6 IPv6 address must use the full 128-bit value.
- If you have upgraded the router from Cisco IOS XR Release 24.3.2 to later releases, one of the four user-configured sessions (created in Cisco IOS XR Release 24.3.x) will be lost, as the router allows only a maximum of three user-configured sessions.

Supported ERSPAN sessions

These tables detail the supported sessions for various releases and SPAN types:

Table 7: Maximum SPAN sessions support on prior and later releases

SPAN type	7.3.1 and prior releases	7.3.2 and later releases
ERSPAN (GRE IPv4, GRE IPv6, or GRE IPv4 + GRE IPv6)	4	8
Local SPAN	4	4
SPAN to File	4	4
Combined SPAN (GRE IPv4 + GRE IPv6 + Local SPAN + SPAN to File)	4	8

Table 8: Maximum sessions supported on various SPAN types

Starting Cisco IOS XR release	SPAN type	Maximum supported sessions
Release 7.2.12	ERSPAN (GRE IPv4, GRE IPv6, or GRE IPv4 + GRE IPv6)	4
	Local SPAN	4
	SPAN to File	4
	Combined SPAN (GRE IPv4 + GRE IPv6 + Local SPAN + SPAN to File)	4
Release 24.2.11 Details sessions supported on the 88-LC1-52Y8H-EM and 88-LC1-12TH24FH-E line cards on all Egress Traffic Management (ETM)-based platforms, when the NPU compatibility mode is set to P100.	ERSPAN (GRE IPv4, GRE IPv6, or GRE IPv4 + GRE IPv6)	4
	Local SPAN	4
	SPAN to File	4
	Combined SPAN (GRE IPv4 + GRE IPv6 + Local SPAN + SPAN to File)	4

For more information on NPU compatibility mode, see [Configure the Compatibility Mode](#).

ERSPAN over GRE

The ERSPAN over GRE IPv6 is a traffic mirroring feature that

- enables mirroring IPv4 or IPv6 traffic in your network
- uses GRE IPv6 for secure encapsulation of mirrored data, and
- sends mirrored traffic to remote analyzers for detailed examination.

Configuration guidelines for ERSPAN over GRE

These guidelines apply to ERSPAN over GRE:

Configuration commands use

Use the **cef proactive-arp-nd enable** command to configure missing adjacency information for the next hop.

How ERSPAN over GRE works

Summary

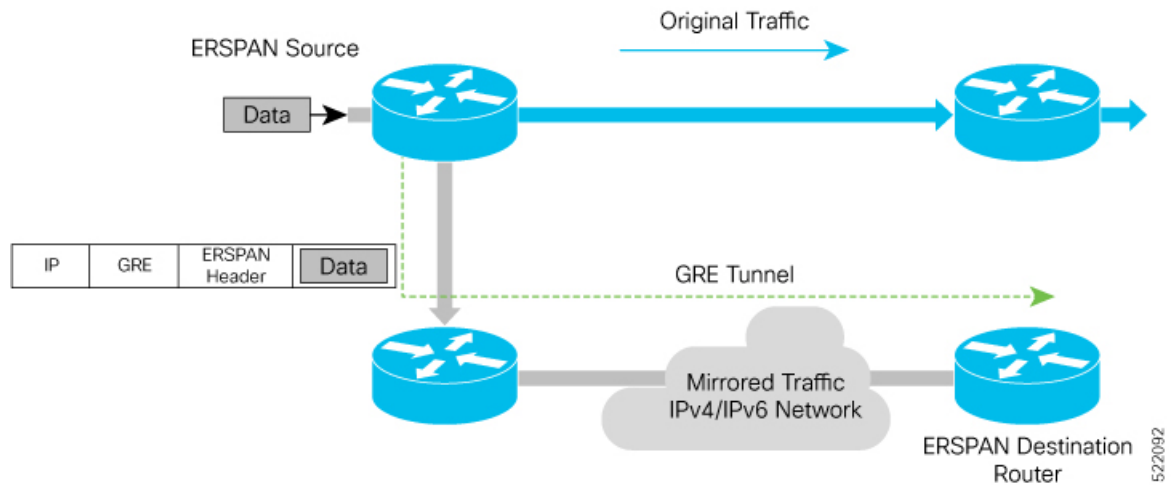
The router encapsulates and mirrors traffic using ERSPAN over GRE IPv6 with the sequence number set to 0, sends it to a remote analyzer, and resolves the next-hop address for successful delivery.

The key components involved in the process are:

- ERSPAN header: Used to encapsulate the mirrored traffic.
- GRE IPv6 packet: Provides the tunneling mechanism for carrying ERSPAN traffic.
- Sequence number: Set to 0 in the GRE header for ERSPAN packets.
- Remote Traffic Analyzer: Receives and monitors the mirrored traffic.
- Next-Hop address resolution: Ensures the router can deliver packets by resolving ARP or neighbor information for the GRE IPv6 tunnel.

Workflow

Figure 3: ERSPAN over GRE



These are the stages of how ERSPAN over GRE works:

1. The router encapsulates the traffic by adding an ERSPAN header inside the GRE IPv6 packet. The GRE header of the ERSPAN encapsulated packets has the sequence number set to 0.
2. The router sends the replicated traffic packet to the destination for monitoring through the GRE IPv6 channel for traffic mirroring.
3. The router sends the mirrored traffic to a remote traffic analyzer for monitoring.
4. The router must resolve the ARP or neighbor for ERSPAN GRE IPv6 tunnel next-hop. We recommend using the **cef proactive-arp-nd enable** command to configure missing adjacency information for the next hop.

Configure ERSPAN over GRE

Use these steps to configure ERSPAN over GRE IPv6:

Procedure

Step 1 Enable GRE IPv6 tunnel configuration.

Example:

```
Router#configure
Router(config)#interface tunnel-ip1
Router(config-if)#tunnel mode gre ipv6
Router(config-if)#tunnel source 2001:DB8:1::1
Router(config-if)#tunnel destination 2001:DB8:2::1
Router(config-if)#no shut
Router(config)#commit
```

A GRE IPv6 tunnel is configured on an interface.

Step 2 Enable an ERSPAN session.

Example:

```
Router#configure
Router(config)#monitor-session mon1 ethernet
Router(config-mon)#destination interface tunnel-ip1
Router(config-mon)#commit
Router(config-mon)#end
```

Step 3 Configure the ERSPAN session under the port to be monitored.

Example:

```
Router(config)#interface HundredGigE0/1/0/14
Router(config-if)#monitor-session mon1 ethernet direction rx-only
Router(config-if-mon)#exit
Router(config-if)#exit
Router(config)#interface Bundle-Ether1
Router(config-if)#monitor-session mon1 ethernet direction rx-only
Router(config-if-mon)#exit
Router(config-if)#exit
Router(config)#interface HundredGigE0/1/0/15.100
Router(config-subif)#monitor-session mon1 ethernet direction rx-only
```

Step 4 Verify the configuration of the ERSPAN over GRE IPv6 feature using the show monitor-session status command.

```
Router#show monitor-session mon1 status
Monitor-session mon1
Destination interface tunnel-ip1
```

```
=====
Source Interface          Dir      Status
-----
Hu0/1/0/14                Rx      Operational
Hu0/1/0/15.100            Rx      Operational
BE1                       Rx      Operational
BE1.1                     Rx      Operational
```

```
Router#show monitor-session erspan3 status internal
Thu Jul 15 06:00:14.720 UTC
Information from SPAN Manager and MA on all nodes:
Monitor-session erspan3 (ID 0x00000007) (Ethernet)
SPAN Mgr: Destination interface tunnel-ip372 (0x0f00049c)
Last error: Success
Tunnel data:
  Mode: GREoIPv6
  Source IP: 77:3:1::79
  Dest IP: 95::90
  VRF:
  ToS: 100
  TTL: 200
  DFbit: Not set
0/3/CPU0: Destination interface tunnel-ip372 (0x0f00049c)
Tunnel data:
  Mode: GREoIPv6
  Source IP: 77:3:1::79
  Dest IP: 95::90
  VRF:
  ToS: 100
  TTL: 200
  DFbit: Not set
0/RP0/CPU0: Destination interface tunnel-ip372 (0x0f00049c)
Tunnel data:
  Mode: GREoIPv6
```

```

Source IP: 77:3:1::79
Dest IP: 95::90
VRF:
ToS: 100
TTL: 200
DFbit: Not set
Information from SPAN EA on all nodes:
Monitor-session 0x00000007 (Ethernet)
0/3/CPU0: Name 'erspan3', destination interface tunnel-ip372 (0x0f00049c)
Platform, 0/3/CPU0:
Monitor Session ID: 7

Monitor Session Packets: 2427313444
Monitor Session Bytes: 480591627492

```

Partial packet capture for ERSPAN

Partial packet capture is a traffic mirroring mechanism that:

- allows ERSPAN to capture only a portion of incoming packets (Rx), rather than capturing the entire packet, and
- allows you to focus on specific header information without needing the entire payload

Benefits of partial packet capture

These are the benefits of capturing partial packets for ERSPAN:

- With less data to process, network analysis and troubleshooting can be faster, allowing quicker identification of issues.
- By capturing only part of the packet, you reduce the amount of data that needs to be mirrored for monitoring. This can alleviate bandwidth consumption and memory requirements.

Configuration guidelines for partial packet capture

These are the guidelines for configuring partial packet capture:

- Enables the new ERSPAN GREv4 and GREv6 truncation configuration per device.
- Configures truncation settings specifically on monitor sessions. Packets received from all sources are truncated if truncation is enabled on a monitor session.
- By default, the whole packet is mirrored without the **mirror first number** (truncation size) configuration.
- If the truncation size configured for a monitor session is less than 343 bytes, the entire packet is mirrored instead of being truncated.
- If the truncation size configured for a monitor session exceeds 343 bytes, the configuration is accepted; however, only 343 bytes will be truncated as this is the maximum supported size.
- Displays an informative system message, **ios-msg**, to warn the user when the configured truncation size exceeds 343 bytes. Example message:

ERSPAN only supports a 343-byte truncation size. The monitor session with session ID <id> will be set to 343 bytes only.

Restrictions for partial packet capture

These restrictions apply to partial packet capture:

- Partial packet capture or packet truncation is per monitor session.
- Truncation per interface is not supported.
- Packet truncation applies for SPAN to file and ERSPAN interfaces only. If you change the destination to local SPAN, then an **ios_msg** is displayed as a warning. The entire packet is mirrored after this message is displayed.

ios_msg example:

The Partial Packet Capture feature is not supported by Local SPAN. The entire Packet will be mirrored.

Configure partial packet capture for ERSPAN

Use this procedure to configure partial packet capturing for ERSPAN on incoming (Rx) traffic.

Procedure

- Step 1** Create an ERSPAN monitor session for mirroring the packets to the tunnel-ip 30 with truncation enabled, and use the **mirror first** option to enable partial packet capturing.

Example:

```
monitor-session mon1 ethernet
 destination interface tunnel-ip 30
 mirror first 343
!
```

The **mon1** monitoring session sets the size of truncation packets for an ERSPAN session to 343.

- Step 2** Attach the session to interfaces and specify the **direction** as **rx-only**.

Example:

```
interface HundredGigE0/0/0/12
 monitor-session mon1 ethernet direction rx-only
!
```

- Step 3** Verify the configuration by using the **show monitor-session status internal** command.

Example:

```
Router#show monitor-session mon1 status internal
Fri Apr 12 18:50:45.006 UTC
Information from SPAN Manager and MA on all nodes:
Packet truncation size: 343B
Monitor-session mon1 (ID 0x00000001) (Ethernet)
SPAN Mgr: Destination interface Tunnel-IP 20 (0x0f000250)
Last error: Success
```

```
Information from SPAN EA on all nodes:  
Monitor-session 0x00000001 (Ethernet)  
0/RP0/CPU0: Name 'mon1', destination interface Tunnel-IP 20 (0x0f000250)  
Platform, 0/RP0/CPU0:
```

```
Monitor Session Packets: 142462
```

```
Monitor Session Bytes: 7653237
```

ERSPAN with flexible CLI

ERSPAN with flexible CLI is an option to configure ERSPAN to

- function as a standalone configuration object that includes all ERSPAN session properties, tunnel properties, and source interfaces
- allow easy removal and re-addition of configurations, and
- minimize user error and support operational simplicity.

Benefits

The flexible CLI for ERSPAN provides benefits, such as:

- Simplifies configuration by consolidating settings into one object.
- Reduces the risk of user errors during configuration changes.
- Enables easy removal and re-application of settings.

ERSPAN rate limit

ERSPAN rate limit is a feature that:

- controls the amount of mirrored traffic sent to an ERSPAN destination over a network.
- enables you to monitor traffic flow through any IP network, including third-party switches and routers.
- prevents network congestion, and
- ensures ERSPAN traffic avoids overloading the network infrastructure.

Table 9: Feature History Table

Feature Name	Release Information	Description
ERSPAN rate limit	Release 25.1.1	Introduced in this release on: Fixed Systems (8700 [ASIC: K100], 8010 [ASIC: A100]) This feature is now supported on: • 8712-MOD-M • 8011-4G24Y4H-I
ERSPAN rate limit	Release 24.4.1	Introduced in this release on: Fixed Systems(8200, 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*). This feature helps you monitor traffic flow through any IP network including third-party switches and routers by providing rate limiting of the mirroring traffic. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 8712-MOD-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH

Benefits

The ERSPAN rate limit feature offers these advantages:

- Limits mirrored traffic to prevent network congestion.
- Enables the use of mirrored traffic for data analysis without overwhelming resources.
- Supports traffic monitoring across various network devices.

Configuration guidelines and restrictions for ERSPAN rate limit

ERSPAN rate limit configuration guidelines

Consider these guidelines for using ERSPAN rate limit effectively:

- Configure the rate limit to prevent exceeding network capacity.
- Monitor traffic flow through any IP network to ensure compatibility with third-party devices.

- Set Quality of Service (QoS) parameters, Traffic Class (0 to 7), on the traffic monitor session to prioritize traffic.



Note Traffic class 0 has the lowest priority and class 7 has the highest. The default traffic class remains the same as the original traffic class.

- Intermediate switches carrying ERSPAN traffic from source session to termination session can belong to any Layer 3 network.
- The ERSPAN rate limit feature is applied on router interfaces to manage monitored traffic.

ERSPAN rate limit restrictions

These restrictions apply to ERSPAN rate limit:

- Exceeding the rate limit may result in the router capping or dropping monitored traffic.
- ERSPAN operates in source and destination sessions, requiring careful configuration to avoid data loss.

How ERSPAN rate limit works

Summary

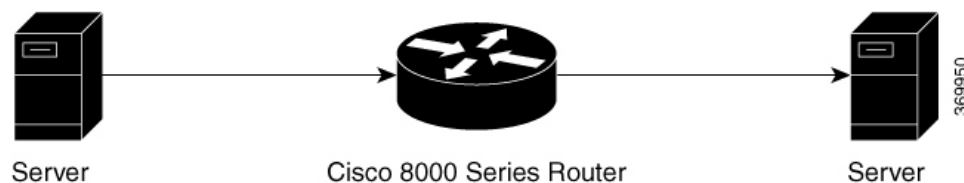
ERSPAN rate limit operates by encapsulating packets in ARPA or IP format using GRE encapsulation.

The key components involved in ERSPAN are:

- ERSPAN source session: The box where the traffic originates or is SPANned.
- ERSPAN termination session or destination session: The box where the traffic is analyzed.

Workflow

Figure 4: Topology for ERSPAN rate limit



These stages describe how ERSPAN rate limit works:

- The router sends GRE tunneled packets to a destination that is identified by an IP address.
- At the destination, SPAN-ASIC decodes and forwards the packets through a port.
- The router analyzes the traffic received at the destination.

Result

The ERSPAN rate limit process results in efficiently managing network traffic by controlling the amount of mirrored traffic sent to a destination.

Configure ERSPAN rate limit

Use this procedure to configure the rate limit for ERSPAN.

Procedure

Step 1 Start a monitor session by using the **monitor-session ERSPAN ethernet** command.

Example:

```
Router# configure
Router(config-if)# monitor-session ERSPAN ethernet
!
```

Step 2 Set the destination interface.

Example:

```
Router(config-if)# destination interface tunnel-ip100
```

Sets the destination interface as **tunnel-ip100**.

Step 3 Configure the interface with the necessary IP address and tunnel mode.

Example:

```
Router(config-if)# interface tunnel-ip1
Router(config-if)# ipv4 address 4.4.4.1 255.255.255.0
Router(config-if)# tunnel mode gre ipv4
Router(config-if)# source 20.1.1.1
Router(config-if)# tunnel destination 20.1.1.2
```

Step 4 Install the policy-map on the appropriate interface to manage traffic in the ingress direction.

Example:

```
Router# configure
Router(config)# interface HundredGigE0/0/0/16
Router(config-if)# ipv4 address 4.4.4.1 255.255.255.0
Router(config-if)# ipv6 address 3001::2/64
Router(config-if)# monitor-session ERSPAN ethernet direction rx-only port-level
acl
!
```

Step 5 Verify the configuration.

Example:

```
Router#show monitor-session FOO status detail
Monitor-session FOO
Destination interface tunnel-ip100
Source Interfaces
-----
TenGigE0/6/0/4/0
Direction: Both
```

Port level: True
ACL match: Disabled

ERSPAN rate-limit per destination

ERSPAN rate-limit per destination is an ERSPAN (Encapsulated Remote SPAN) feature that controls the amount of mirrored traffic sent to ERSPAN destinations by applying a rate-limit per source NPU.

Table 10: Feature History Table

Feature Name	Release Information	Description
ERSPAN rate-limit per destination	Release 25.3.1	Introduced in this release on: Fixed Systems (8200 [ASIC: Q100, Q200, P100], 8700 [ASIC: P100, K100], 8010 [ASIC: A100]); Centralized Systems (8600 [ASIC: Q200]); Modular Systems (8800 [LC ASIC: Q100, Q200, P100]) You can now control the amount of mirrored traffic sent to ERSPAN destinations by applying a rate-limit per source NPU. This helps prevent network congestion and optimize resource usage. The feature introduces these changes. CLI: The rate-limit value keyword is introduced in the monitor-session command. Yang Data Model: • New Xpaths for <code>Cisco-IOS-XR-um-monitor-session-cfg.yang</code> • New Xpaths for <code>Cisco-IOS-XR-Ethernet-SPAN-cfg.yang</code>

Benefits

The ERSPAN rate-limit per destination feature offers these advantages:

- Allows you to apply rate-limiting directly to SPAN mirrored traffic.
- Streamlines configuration with a single command by using **monitor-session session-name rate-limit value** command.
- Provides enhanced control over the amount of mirrored traffic.
- Improves monitoring capabilities by delivering per-session statistics.
- Simplifies tracking of mirrored traffic and aiding troubleshooting of network monitoring issues.

Configuration guidelines for ERSPAN rate-limit per destination

ERSPAN rate-limit per destination configuration guidelines

- You can only apply the rate-limit to ERSPAN destinations. These destinations use tunnel interfaces. The system rejects configurations for local SPAN, which uses physical interfaces.
- Cisco IOS XR Release 25.3.1 supports rate-limit in kilobits per second (kbps). The supported values range from 40000 kbps (40 Mbps) as the minimum to 40000000 kbps (40 Gbps) as the maximum.
- The system applies the configured rate-limit per ingress NPU. An NPU can host multiple source interfaces.
- The effective rate-limit scales with the number of NPUs that contain source interfaces for a given session. This means the actual rate-limit can be higher than the configured value.

For example, if you configure 200 kbps on a system with 48 NPUs:

- If only one NPU has SPAN source interfaces, the rate-limit is 200 kbps.
 - If two NPUs each have SPAN source interfaces, the rate-limit is 400 kbps.
 - If all 48 NPUs have SPAN source interfaces, the rate-limit scales to 9600 kbps.
- Do not configure rate-limiting with **drops** or **protocol-capture** options. The system rejects such configurations.

Restrictions for ERSPAN rate-limit per destination

ERSPAN rate-limit per destination restrictions

- Rate-limit applies only to ERSPAN (mirroring over GRE tunnel), not supported for local SPAN (mirroring over a physical interface).
- Rate-limit configuration is supported only for tunnel interfaces and rejected for Local SPAN (non-tunnel interfaces).
- Rate-limiting applies only to transient (forwarded) traffic. It does not support packet drops, for-us packets, or locally sourced packets.
- The system provides per-session counters. It does not provide per-interface or per-direction statistics.

Configure ERSPAN rate-limit per destination

Use this procedure to configure the rate-limit per destination for ERSPAN.

Procedure

- Step 1** Start a monitor session by using the **monitor-session ERSPAN ethernet** command.

Example:

```
Router# configure
Router(config-if)# monitor-session ERSPAN ethernet
!
```

Step 2 Apply the rate-limit for the selected destination.

Example:

```
Router(config-if)# destination interface tunnel-ip100 rate-limit 200kpbs
```

The rate-limit of **200kpbs** is applied for the destination interface **tunnel-ip100**.

Step 3 Verify the configuration.

Example:

```
Router# show monitor-session ERSPAN ethernet status detail
Monitor-session FOO
Destination interface tunnel-ip100
rate-limit: 200kpbs per NPU
Source Interfaces
-----
TenGigE0/6/0/4/0
Direction: Both
Port level: True
ACL match: Disabled
```

Step 4 To display statistics related information, such as the source interfaces and the replicated packet statistics for that interface, use the **show monitor-session counters per-session** command.

Example:

```
Router# show monitor-session foo counters per-session
Monitor session foo
  GigabitEthernet 0/3/0/0.100:
    Rx Replicated: 100 Packets 8000 Bytes
    Tx Replicated: 2 Packets 3000 Bytes
    Non Replicated: 0 Packets 0 Bytes
Per-session counters:
  Monitor-session foo
    Total replicated: 45 packets, 3600 octets
    Non-replicated: 0 packets, 0 octets
```

Note

With a rate-limit applied, counters are available at a session level, rather than per source interface.

Traffic mirroring with DSCP

Traffic mirroring with Differentiated Service Code Point (DSCP) is a traffic mirroring feature that:

- uses the DSCP value in the Differentiated Services (DS) field of an IP packet to classify network traffic
- sets the DSCP value in the six most significant bits of the DS field in the IP header, allowing for 64 different values ranging from 0 to 63
- these six bits affect the Per Hop Behavior (PHB) and determine how a packet is moved forward, and
- places packets into limited traffic classes for prioritization by routers.



Note The DS field was formerly known as Type of Service (ToS).

The table summarizes the service class names as defined in [RFC 2474](#).

Table 11: DSCP, DS, and ToS values

DSCPValue in Decimal	DS Binary	DSHex	DSCPName	DS/ToSValue	ServiceClass
0	000000	0x00	DF/CS0	0	Standard
-	-	-	none	2	
1	000001	0x01	None	4	
1	000001	0x01	LE	4	Lower-effort
2	000010	0x02	None	8	
4	000100	0x04	None	16	
8	001 000	0x08	CS1	32	Low-priority data
10	001 010	0x0a	AF11	40	High-throughput data
12	001 100	0x0c	AF12	48	High-throughput data
14	001 110	0x0e	AF13	56	High-throughput data
16	010 000	0x10	CS2	64	OAM
18	010 010	0x12	AF21	72	Low-latency data
20	010 100	0x14	AF22	80	Low-latency data
22	010 010	0x16	AF23	88	Low-latency data
24	011 000	0x18	CS3	96	Broadcast video
26	011 000	0x1a	AF31	104	Multimedia streaming
28	011 100	0x1c	AF32	112	Multimedia streaming
30	011 110	0x1e	AF33	120	Multimedia streaming
32	100 000	0x20	CS4	128	Real-time interactive
34	100 010	0x22	AF41	136	Multimedia conferencing
36	100 100	0x24	AF42	144	Multimedia conferencing
38	100 110	0x26	AF43	152	Multimedia conferencing
40	101 000	0x28	CS5	160	Signaling(IP telephony, etc)
44	101 100	0x2c	Voice-admit	176	
46	101 110	0x2e	EF	184	Telephony
48	110 000	0x30	CS6	192	Network routing control

56	111 000	0x38	CS7	224	“reserved”
----	---------	------	-----	-----	------------

Benefits of traffic mirroring with DSCP

Benefits of using Traffic mirroring with DSCP are:

- It classifies and manages network traffic.
- It assigns different priority levels to packets.
- It enhances the Quality of Service (QoS) for mirrored packets.

DSCP marking on egress GRE tunnel in ERSPAN

Differentiated Service Code Point (DSCP) marking on egress GRE tunnel in ERSPAN is a mechanism that:

- classifies and manages network traffic by assigning different priority levels to packets, and
- defines the Quality of Service (QoS) for the mirrored packets by configuring the DSCP marking on an egress GRE tunnel for ERSPAN traffic.

Starting with Cisco IOS XR Software Release 7.5.4, you can set or modify the DSCP marking on ERSPAN GRE tunnels. This allows captured traffic routing using GRE encapsulation.

Table 12: Feature History Table

Feature Name	Release Information	Feature Description
DSCP marking on egress GRE tunnel in ERSPAN	Release 25.1.1	Introduced in this release on: Fixed Systems (8700 [ASIC: K100], 8010 [ASIC: A100]) This feature is now supported on: • 8712-MOD-M • 8011-4G24Y4H-I

Feature Name	Release Information	Feature Description
DSCP marking on egress GRE tunnel in ERSPAN	Release 24.4.1	Introduced in this release on: Fixed Systems(8200, 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*). This feature which allows you to control the QoS for your network's ERSPAN GRE tunnel traffic and eases the effort to control your customers' bandwidth across next-hop routers is supported on the following hardware. *This feature is now supported on: • 8212-48FH-M • 8711-32FH-M • 8712-MOD-M • 88-LC1-12TH24FH-E • 88-LC1-52Y8H-EM • 88-LC1-36EH
DSCP marking on egress GRE tunnel in ERSPAN	Release 7.5.4	You can now set or modify Differentiated Service Code Point (DSCP) value on the ERSPAN GRE tunnel header. This feature allows you to control the QoS for your network's ERSPAN GRE tunnel traffic and eases the effort to control your customers' bandwidth across next-hop routers.

Configure DSCP marking on egress GRE tunnel in ERSPAN

Follow this procedure to configure DSCP marking on egress GRE tunnel in ERSPAN. You can configure DSCP value on both IPv4 and IPv6 headers.

Procedure

- Step 1** Enter the terminal configuration mode.
- Example:**
- ```
Router#configure terminal
```
- Step 2** Specify the tunnel interface and configure the DSCP value.

**Example:**

```
Router(config)#interface tunnel-ip1
Router(config-if)#tunnel tos 96
Router(config-if)#tunnel mode gre ipv4
Router(config-if)#tunnel source 192.0.2.1
Router(config-if)#tunnel destination 192.0.2.254
Router(config-if)#commit
```

The router configures the DSCP or ToS value as **96** on the **tunnel-ip1** interface.

**Step 3**

Verify the configuration.

**Example:**

```
Router#show run interface tunnel-ip1
interface tunnel-ip1
ipv4 address 192.0.2.0/24
tunnel tos 96
tunnel mode gre ipv4
tunnel source 192.0.2.1
tunnel vrf red
tunnel destination 192.0.2.254
Router#show monitor-session ERSPAN-2 status internal
Information from SPAN Manager and MA on all nodes:
Monitor-session ERSPAN-2 (ID 0x00000003) (Ethernet)
SPAN Mgr: Destination interface tunnel-ip1 (0x20008024)
Last error: Success
Tunnel data:
Mode: GREoIPv4
Source IP: 192.0.2.1
Dest IP: 192.0.2.254
VRF: red
VRF TBL ID: 0
ToS: 96
TTL: 255
DFbit: Not set
```

This setup confirms the DSCP marking and ensures the correct configuration and operation of traffic mirroring with DSCP.

## DSCP bitmask to filter ingress ERSPAN traffic

DSCP bitmask to filter ingress ERSPAN traffic is a mechanism that:

- filters ingress ERSPAN traffic with a specific DSCP value
- matches the bitmask in the Access Control List (ACL) rule with the DSCP field in the IP packet header, and
- determines if the packet aligns with the desired bitmask, classifying and prioritizing traffic as it enters the network.

Table 13: Feature History Table

| Feature Name                                  | Release Information | Feature Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DSCP bitmask to filter ingress ERSPAN traffic | Release 25.1.1      | <p>Introduced in this release on: Fixed Systems ( 8700 [ASIC: K100], 8010 [ASIC: A100])</p> <p>This feature is now supported on:</p> <ul style="list-style-type: none"> <li>• 8712-MOD-M</li> <li>• 8011-4G24Y4H-I</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| DSCP bitmask to filter ingress ERSPAN traffic | Release 24.4.1      | <p>Introduced in this release on: Fixed Systems(8200, 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*).</p> <p>This feature now allows to mirror multiple traffic flows for matched DSCP value of IP header on the ERSPAN on the following hardware.</p> <p>*This feature is now supported on:</p> <ul style="list-style-type: none"> <li>• 8212-48FH-M</li> <li>• 8711-32FH-M</li> <li>• 8712-MOD-M</li> <li>• 88-LC1-12TH24FH-E</li> <li>• 88-LC1-52Y8H-EM</li> <li>• 88-LC1-36EH</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                     |
| DSCP bitmask to filter ingress ERSPAN traffic | Release 7.5.4       | <p>You can now mirror multiple traffic flows for matched Differentiated Service Code Point (DSCP) value of IP header on the Encapsulated remote SPAN (ERSPAN). The matched DSCP value is based on the DSCP value and the bitmask configured in Access Control List (ACL) rule.</p> <p>Earlier, you could monitor single traffic flow by setting the <a href="#">RFC 4594</a> defined DSCP values in the GRE tunnel header.</p> <p>This feature introduces the following changes:</p> <ul style="list-style-type: none"> <li>• <b>CLI:</b> <a href="#">deny (IPv4)</a>, <a href="#">deny (IPv6)</a>, <a href="#">permit (IPv4)</a>, and <a href="#">permit (IPv6)</a> are modified to include new keyword <b>bitmask</b>.</li> <li>• <b>YANG DATA Model:</b> New XPathS for Cisco-IOS-XR-um-ipv4-access-list-cfg and Cisco-IOS-XR-um-ipv6-access-list-cfg (see <a href="#">Github</a>, <a href="#">YANG Data Models Navigator</a>).</li> </ul> |

### Benefits of using DSCP bitmask to filter ingress ERSPAN traffic (Reference)

Using a DSCP bitmask to filter ingress ERSPAN traffic offers these advantages:

- It allows for specific traffic flow mirroring.
- It enhances traffic classification and prioritization.
- It reduces unnecessary traffic mirroring on incoming ports.

## Configuration guidelines for DSCP bitmask to filter ingress ERSPAN traffic

Use these guidelines to configure the DSCP bitmask to filter ingress ERSPAN traffic:

- Starting , you can configure an ACL rule with DSCP bitmask on the ERSPAN GRE tunnels to mirror specific traffic flows.
- When you configure an ACL with DSCP and DSCP mask on ERSPAN, ERSPAN mirrors the traffic whose DSCP value lies within the combination of DSCP value and the specified mask.
- Without ACL rule, ERSPAN mirrors all the traffic on the incoming port.
- Verify that the DSCP value lies within the specified mask range.
- The router maps the DSCP value to a single traffic class according to the values defined in [RFC2474](#).
- Masking the DSCP value in ACL rule allows mirroring multiple traffic flows.
- DSCP value and mask operate similarly to IPv4 address and mask.

## Configure DSCP bitmask to filter ingress ERSPAN traffic

To configure DSCP bitmask, use the **bitmask** option along with the **dscp** option while configuring the ACL.

Use these steps to configure DSCP bitmask on ingress ERSPAN for IPv4 traffic.

### Procedure

**Step 1** Configure an ACL.

**Example:**

```
Router# config
Router(config)# ipv4 access-list acl1
Router(config-ipv4-acl)# 10 permit ipv4 host 192.0.2.1 any dscp af22 bitmask 0x3f
Router(config-ipv4-acl)# commit
Router(config-ipv4-acl)# exit
```

**Step 2** Attach the created ACL to an interface.

**Example:**

```
Router(config)# interface HundredGigE0/0/0/6
Router(config-if)# ipv4 address 192.0.2.51 255.255.255.0
```

**Step 3** Monitor the ingress ACL applied and DSCP-masked IPv4 traffic on ERSPAN.

**Example:**

```
Router(config-if)# monitor-session TEST ethernet direction rx-only port-level acl ipv4 acl1
Router(config-if)# commit
```

## Monitor ERSPAN sessions with SPAN and security ACLs

Monitoring multiple ERSPAN sessions is a functionality that:

- uses GREv4 and GREv6 under the same source interface to monitor multiple ERSPAN sessions
- allows you to choose the destination interface for the mirrored traffic from the multiple ERSPAN monitor sessions configured on an interface, and
- uses SPAN and security ACLs together for the configuration of monitor sessions.

**Table 14: Feature History Table**

| Feature Name                                                | Release Information | Feature Description                                                                                                                                                                                                |
|-------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monitor multiple ERSPAN sessions with SPAN and security ACL | Release 25.1.1      | Introduced in this release on: Fixed Systems (8700 [ASIC: K100], 8010 [ASIC: A100])<br><br>This feature is now supported on: <ul style="list-style-type: none"><li>• 8712-MOD-M</li><li>• 8011-4G24Y4H-I</li></ul> |

| Feature Name                                                | Release Information | Feature Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monitor multiple ERSPAN sessions with SPAN and security ACL | Release 24.4.1      | <p>Introduced in this release on: Fixed Systems(8200, 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*).</p> <p>This feature now enables you to use SPAN and security ACL together to monitor multiple ERSPAN sessions under the same source interface thus distributing the mirrored traffic over different destination interfaces and allowing selective incoming traffic on the following hardware.</p> <p>*This feature is now supported on:</p> <ul style="list-style-type: none"> <li>• 8212-48FH-M</li> <li>• 8711-32FH-M</li> <li>• 8712-MOD-M</li> <li>• 88-LC1-12TH24FH-E</li> <li>• 88-LC1-52Y8H-EM</li> <li>• 88-LC1-36EH</li> </ul> |
| Monitor multiple ERSPAN sessions with SPAN and security ACL | Release 7.5.4       | <p>With this feature, you can use SPAN and security ACL together to monitor multiple ERSPAN sessions under the same source interface. SPAN ACL helps you to distribute the mirrored traffic over different destination interfaces and Security ACL helps you to allow selective incoming traffic.</p>                                                                                                                                                                                                                                                                                                                                                                                      |

## Configure multiple monitor ERSPAN sessions with SPAN and security ACL

Use this procedure to configure SPAN and security ACL with GREv4 and GREv6 monitor sessions.

### Procedure

**Step 1** Start a monitor session and attach the SPAN ACL to an interface.

**Example:**

```

Router# configure
Router(config-if)#monitor-session always-on-v4
Router(config-if)#monitor-session always-on-v4 ethernet direction rx-only port-level
Router(config-if-mon)#acl ipv4 v4-monitor-acl1
Router(config-if-mon)#acl ipv6 v6-monitor-acl1
Router(config-if-mon)#exit
Router(config-if)#monitor-session on-demand-v4 ethernet direction rx-only port-level
Router(config-if-mon)#acl ipv4 v4-monitor-acl2
Router(config-if-mon)#acl ipv6 v6-monitor-acl2
Router(config-if-mon)#exit

```

**Step 2** Attach the security ACL to an interface.

**Example:**

```

Router(config-if)#ipv4 access-group sec_aclv4 ingress
Router(config-if)#ipv6 access-group sec_aclv6 ingress
Router(config-if)#commit

```

## Selective packet capture for protocol and drops

Selective packet capture is a traffic mirroring feature that

- enables you to select specific network packets for monitoring or analysis,
- allows filtering based on Tx and Rx directions and protocol-specific attributes, and
- supports expressions in a standard format similar to the tcpdump tool.

| Feature Name                                    | Release Information | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Selective packet capture for protocol and drops | Release 25.4.1      | <p>Introduced in this release on: Fixed Systems (8200 [ASIC: Q100, Q200, P100], 8700 [ASIC: P100, K100], 8010 [ASIC: A100]); Centralized Systems (8600 [ASIC: Q200]); Modular Systems (8800 [LC ASIC: Q100, Q200, P100])</p> <p>Gain enhanced control over mirrored traffic by filtering and capturing specific dropped and protocol packets. Apply filters during monitor session setup to improve SPAN packet capture. This approach leads to more effective network troubleshooting.</p> <p>You can filter packets by direction (incoming, outgoing, or both) using pcap filter expressions to target packets of interest. Filters help reduce captured data and focus on relevant packets.</p> <p>The feature introduces these changes.</p> <p>CLI:</p> <ul style="list-style-type: none"> <li>• The <b>protocol-capture</b> and <b>filter expression</b> keywords are introduced in the <b>monitor-session</b> command.</li> <li>• The <b>filter expression</b> keyword is introduced in the <b>drops</b> command.</li> </ul> |

Selective packet capture is applied to both dropped and protocol packets within SPAN monitor sessions by using the **monitor-session** command. You can configure these filters separately for dropped packets or protocol packets in each session. Filters help reduce the amount of captured data and target specific packets, such as those from a particular source IP address.

### Benefits

Selective packet capture offers these benefits:

- Increases efficiency by reducing the volume of mirrored traffic through granular filtering.
- Allows you to capture specific dropped packets and protocol packets.
- Diagnoses network issues more effectively.

## Configuration guidelines for selective packet capture

Consider these guidelines for using Selective packet capture effectively.

- Use valid **pcap** filter expressions to filter the captured packets.
- Configure **protocol-capture** or **drops** only on supported destination types such as file, sFlow, or GRE tunnel.
- Remove any incorrect attachments when setting up protocol-capture, as attachments are not supported in protocol-capture sessions.

## Restrictions for Selective packet capture

These restrictions apply to Selective packet capture.

- Drops and protocol-capture cannot be configured simultaneously on the same session.
- For ERSPAN, rate-limit features cannot be used in conjunction with drops or protocol-capture configuration.
- IPv4 broadcast address filtering is not supported.
- The number of global drop and protocol capture sessions is limited.
- Any attachments to monitor sessions with protocol-capture configuration are disabled and will show as errors in **show monitor-session** commands.

## Configure Selective packet capture for protocol and drops

Use this task to capture drop or protocol packets by using SPAN sessions, applying direction, and content-based filters. This allows more targeted traffic analysis and efficient troubleshooting.

### Before you begin

- Verify that the router supports protocol-capture and direction-based filtering.
- Ensure that the destination of monitor session is set to file, sFlow, or ERSPAN.



- Prepare valid pcap filter expressions if advanced filtering is required.
- Confirm that drops and protocol-capture are not configured in the same monitor session.

Follow these steps to configure Selective packet capture for protocol and drops:

## Procedure

**Step 1** Create or identify a monitor-session for packet capture.

### Example:

```
Router(config)# monitor-session mon1 [ethernet]
```

**Step 2** Set the destination for packet capture to file, sFlow, or an ERSPAN tunnel as supported.

### Example:

```
Router(config-mon)# destination file size 10000 format pcapng
```

**Step 3** Configure selective packet capture for drop packets.

```
monitor-session <name> [ethernet] destination {file | sflow | interface <tunnel>} drops
[packet-processing] [traffic-management] [rx | tx] [filter <expression>]
```

### Example:

Capture drop packets from a particular source IP address.

```
Router(config)# monitor-session span1 ethernet
Router(config-mon)# destination file size 10000 format pcapng
Router(config-mon)# drops packet-processing rx filter ip src host 192.0.2.1
Router(config-mon)# commit
```

**Step 4** (Optional) Validate the configuration for drop packets and check for operational errors or unsupported attachments.

### Example:

```
Router(config)# show monitor-session status detail
```

#### Command Output for drop packets:

```
Thu Jul 31 19:40:32.709 UTC
Monitor-session span2
 Destination File (pcapng) - Not collecting
 All packet-processing drops:
 Direction: Rx
 Drops filtering:
 PCAP filter ip src host 192.0.2.1

Source Interfaces

```

**Step 5** Configure selective packet capture for protocol-capture.

```
monitor-session <name> [ethernet] destination {file | sflow | interface <tunnel>} protocol-capture
[rx | tx] [filter <expression>]
```

### Example:

Capture all protocol packets in the receive (Rx) direction from a specific protocol.

```
Router(config)# monitor-session span2 ethernet
Router(config-mon)# destination file size 10000 format pcapng always-on periodic-write 10
```

```
Router(config-mon)# protocol-capture rx filter arp
Router(config-mon)# commit
```

**Step 6** (Optional) Validate the configuration for protocol-capture and check for operational errors or unsupported attachments.

**Example:**

```
Router(config)# show monitor-session status detail
```

**Command Output for protocol packets:**

```
Monitor-session span2
```

```
Destination File (pcapng) - Packet collecting (always-on)
```

```
Periodic write interval: 10 seconds
```

```
Protocol packets:
```

```
Direction: Rx
```

```
PCAP filter arp
```

```
Source Interfaces
```

```

```

**Step 7** (Optional) Use [show monitor-session counters](#) command to view packet counter statistics and to monitor replicated and non-replicated packets per session.

---

The SPAN session captures and saves packets to the specified destination file, filtered by direction and the defined pcap filter expression. Only packets matching the configuration are mirrored and saved.

**What to do next**

- Review the captured pcap files for analysis.
- Check per-session counters to validate expected capture rates.
- Remove or adjust filters as needed to refine packet capture scope.
- Re-enable interface attachments if protocol-capture configuration is removed.



## CHAPTER 6

# SPAN to file

This chapter explains a traffic monitoring feature that captures traffic to PCAP files for later analysis. It details capabilities such as partial packet capture, transmit (Tx) direction mirroring, continuous Always-On capture with periodic writes, and Unique Capture to reduce redundant data.

- [SPAN to file, on page 57](#)
- [Configure SPAN to file for truncation, on page 66](#)
- [Configure SPAN to file for direction, on page 67](#)
- [Always-On SPAN to file with periodic write, on page 68](#)
- [SPAN to file with unique capture, on page 72](#)

## SPAN to file

SPAN to file is a traffic monitoring feature that

- captures traffic from a SPAN session and writes it directly to a file for later analysis
- extends the existing SPAN feature by mirroring network packets to a file instead of an interface
- helps in the analysis of the packets at a later stage, and
- saves the file in the PCAP file format, which is compatible with tools like tcpdump and Wireshark.

The **monitor-session** *<name>* [**ethernet**|**ipv4**|**ipv6**|**mpls-ipv4**|**mpls-ipv6**] command creates a monitor-session with the specified name and traffic class, serving as a chain point from the SPAN feature.

The **destination file** [**size** *<kbytes>*] [**buffer-type** **linear**] command adds a **file** destination option to the session configuration.

**destination file** has these configuration options:

- Buffer size: Sets the maximum file size for captured packets.
- Buffer types:
  - Circular: This is the default buffer type. Once the buffer is full, it overwrites from the beginning.
  - Linear: Once the buffer is full, no further packets are logged. Only the linear buffer type must be explicitly configured.

Altering any parameters, such as buffer size or type, recreates the session and clears any packet buffers.

All configuration options available for other SPAN types should also be supported by SPAN to file. These include options such as applying ACLs and capturing only the first X bytes of each packet.

Starting with Cisco IOS XR Release 7.5.3, truncation is supported per global session rather than per interface. These options are implemented by the router when punting the packet.

**Table 15: Feature History Table**

| Feature name                     | Release information | Feature description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SPAN-to-File with unique capture | Release 25.2.1      | <p>Introduced in this release on: Fixed Systems (8200 [ASIC: Q200, P100], 8700 [ASIC: P100, K100], 8010 [ASIC: A100]); Centralized Systems (8600 [ASIC: Q200]); Modular Systems (8800 [LC ASIC: Q100, Q200, P100])</p> <p>This feature enhances the SPAN-to-File functionality by allowing you to capture only a single, unique packet for each punt reason or interface. This prevents interesting packets from being overshadowed by repeated packets in the analysed flow, ensuring that diverse and relevant packets are retained for analysis.</p> <p>The feature introduces these changes:</p> <p>CLI:</p> <p>The <b>unique-punt</b> and <b>unique-port</b> keywords are introduced in the <b>drops</b> command.</p> |

| Feature name                               | Release information | Feature description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Always-On SPAN-to-File with periodic write | Release 24.4.1      | <p>Introduced in this release on: Fixed Systems(8200, 8700); Centralized Systems (8600); Modular Systems (8800 [LC ASIC: Q100, Q200, P100]).</p> <p>The routers can now provide reliable, always-available packet capture for post-event analysis, eliminating the need for prior configuration or user interaction.</p> <p>The enhanced SPAN-to-File feature provides continuous packet capture and debugging capability with always-on functionality that starts automatically upon destination configuration. It prevents data loss during node reloads by periodically writing packet buffer contents to disk, without stopping the capture. A default SPAN-to-File session for forwarding and buffer drops is always active and can be disabled if not needed. The feature also supports packet truncation and sampling in software for software-mirrored packets, independent of NPU capabilities.</p> <p>The feature introduces these changes:</p> <p>CLI:</p> <ul style="list-style-type: none"> <li>• <b>monitor-session default-capture-disable</b></li> <li>• <b>monitor-session local-capture-capacity</b></li> <li>• The <b>always-on</b>, <b>periodic-write</b>, and <b>capacity</b> keywords are introduced in the <b>destination file</b> command.</li> <li>• The <b>write</b> keyword is introduced in the <b>monitor-session &lt;name&gt; packet-collection</b> action command.</li> </ul> <p>YANG data models:</p> <ul style="list-style-type: none"> <li>• New Xpaths for <code>Cisco-IOS-XR-um-monitor-session-cfg.yang</code></li> <li>• New Xpaths for <code>Cisco-IOS-XR-Ethernet-SPAN-cfg.yang</code></li> <li>• New Xpaths for <code>Cisco-IOS-XR-Ethernet-SPAN-act.yang</code></li> </ul> <p>(see <a href="#">GitHub</a>, <a href="#">YANG Data Models Navigator</a>)</p> |
| SPAN-to-file in Tx direction               | Release 25.1.1      | <p>Introduced in this release on: Fixed Systems (8700 [ASIC: K100], 8010 [ASIC: A100])</p> <p>This feature is now supported on:</p> <ul style="list-style-type: none"> <li>• 8712-MOD-M</li> <li>• 8011-4G24Y4H-I</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Feature name                                         | Release information | Feature description                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SPAN-to-file in Tx direction                         | Release 24.4.1      | <p>Introduced in this release on: Fixed Systems(8200, 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*).</p> <p>This feature now allows to capture packets in the Tx direction on the following hardware.</p> <p>*This feature is now supported on:</p> <ul style="list-style-type: none"> <li>• 8212-48FH-M</li> <li>• 8711-32FH-M</li> <li>• 8712-MOD-M</li> <li>• 88-LC1-12TH24FH-E</li> <li>• 88-LC1-52Y8H-EM</li> <li>• 88-LC1-36EH</li> </ul> |
| SPAN-to-file support in Tx and Rx direction          | Release 7.5.3       | <p>With this feature, the ability to capture the packet in Tx direction along with the ability to store the capture on the file is supported.</p> <p>You can now capture the packet in the Tx direction and store the capture on the file. Earlier, you could only capture or mirror the traffic in the Rx direction. You now have the flexibility to choose Tx, Rx, or both directions.</p> <p>You can now capture and analyze the outgoing (Tx) packets.</p>                                |
| Partial packet capture ability for SPAN-to-file (Rx) | Release 25.1.1      | <p>Introduced in this release on: Fixed Systems ( 8010 [ASIC: A100])</p> <p>This feature is now supported on:</p> <ul style="list-style-type: none"> <li>• 8011-4G24Y4H-I</li> </ul>                                                                                                                                                                                                                                                                                                          |

| Feature name                                         | Release information | Feature description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Partial packet capture ability for SPAN-to-file (Rx) | Release 24.4.1      | <p>Introduced in this release on: Fixed Systems(8200, 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*).</p> <p>This feature now allows you to perform partial packet capture in the Rx direction on the following hardware.</p> <p>*This feature is now supported on:</p> <ul style="list-style-type: none"> <li>• 8212-48FH-M</li> <li>• 8711-32FH-M</li> <li>• 8712-MOD-M</li> <li>• 88-LC1-12TH24FH-E</li> <li>• 88-LC1-52Y8H-EM</li> <li>• 88-LC1-36EH</li> </ul>                                     |
| Partial packet capture ability for SPAN-to-file (Rx) | Release 7.5.3       | <p>With this feature, you can perform partial packet capture in the Rx direction.</p> <p>Earlier, the ability for entire packet capture was available in the Tx direction only, now you can choose entire or partial packet capture in the Rx direction also.</p> <p>Here, partial packet capture is also known as truncation.</p>                                                                                                                                                                                                                   |
| SPAN-to-file PCAPng file format                      | Release 24.4.1      | <p>Introduced in this release on: Fixed Systems(8200, 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*).</p> <p>This PCAPng File Format feature that contains different blocks used to rebuild the captured packets into recognizable data is now supported on the following hardware.</p> <p>*This feature is now supported on:</p> <ul style="list-style-type: none"> <li>• 8212-48FH-M</li> <li>• 8711-32FH-M</li> <li>• 88-LC1-12TH24FH-E</li> <li>• 88-LC1-52Y8H-EM</li> <li>• 88-LC1-36EH</li> </ul> |

| Feature name                    | Release information | Feature description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SPAN-to-file PCAPng file format | Release 7.3.1       | <p>PCAPng is the next generation of packet capture format that contains a dump of data packets captured over a network and stored in a standard format.</p> <p>The PCAPng file contains different types of information blocks, such as the section header, interface description, enhanced packet, simple packet, name resolution, and interface statistics. These blocks can be used to rebuild the captured packets into recognizable data.</p> <p>The PCAPng file format:</p> <ul style="list-style-type: none"> <li>• Provides the capability to enhance and extend the existing capabilities of data storage over time</li> <li>• Allows you to merge or append data to an existing file.</li> <li>• Enables to read data independently from network, hardware, and operating system of the machine that made the capture.</li> </ul> |

This table details the support for packet capture for various releases:

| Before Release 7.5.3                                                                                                                | After Release 7.5.3                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| There was no functionality that you could enable to capture the payload of packets coming from your customers for security reasons. | The capture of all the outgoing packets from the router is supported. |

## Configuration guidelines for SPAN to file

These guidelines apply to SPAN to file:

- A maximum of 100 source ports are supported across the system. Individual platforms may support lower numbers.
- All the SPAN sessions are configured under the Ethernet class.
- At any given time, the system supports four SPAN to file sessions.
- When you attach multiple interfaces to a monitor session, the minimum buffer size is 1KB. The maximum buffer size is 1000KB, and default buffer size is 2KB.

## Restrictions for SPAN to file

These restrictions apply to SPAN to file:

- Only incoming packet mirroring on the source interface is supported. Outgoing mirrored packets cannot be saved to a file. Starting with Cisco IOS XR Software Release 7.5.3, there are no restrictions.
- You can apply SPAN ACLs only in ingress direction. Hence, ACLs for SPAN to file can only be applied in ingress direction.



- ACL on MPLS traffic is not supported.
- MPLS over GRE traffic is supported, however, GRE interfaces cannot be configured as source interfaces.
- Packet truncation applies for SPAN to file and ERSPAN interfaces only. If you change the destination to local SPAN, then an **ios\_msg** is displayed as a warning. The entire packet is mirrored after this message is displayed.

**ios\_msg** example:

```
The Partial Packet Capture feature is not supported by Local SPAN. The entire
Packet will be mirrored.
```

- For outgoing (Tx) SPAN to file, security ACL is not supported.
- For outgoing (Tx) SPAN to file, only transit traffic is mirrored.
- Self-originating traffic cannot be mirrored.

## Capabilities of SPAN to file

These are the supported capabilities of SPAN to file:

### Traffic types and interface support

- Mirror outgoing traffic and punt it to the CPU across all NPU versions.
- Mirror outgoing IPv4, IPv6, and MPLS traffic to file.
- Mirror outgoing traffic across all types of L3 interfaces, including physical, subinterfaces, bundle, and bundle sub-interfaces.
- Mirror outgoing traffic across L2 or BVI interfaces.

### Direction and mirroring control

Enable SPAN to File truncation configuration for both RX and TX directions. You can specify the **both** keyword to enable RX and TX mirroring on a single source interface.

### Truncation configuration

- Configure a different truncation size on each monitor session.
- Configure SPAN to file mirroring packet truncation size from 1 to 10,000 bytes. Values outside this range are rejected with an error.
- Change the truncation size when packet collection has stopped; removing or re-adding the monitor session is not required.
- Change the truncation size during packet collection without stopping the monitor session.
- Mirror the entire packet by default if no truncation size is configured. If the packet size is smaller than the truncation size, the entire packet is mirrored.

## How SPAN to file works

### Summary

SPAN to file works by capturing traffic from a SPAN session and saving the data to a file for later analysis.

### Workflow

These stages describe how SPAN to file works when you configure a single file as a destination for a SPAN session:

- The system creates a buffer on each node, where the network packets are logged.
- The system creates buffer for all packets on the node regardless of which interface they are from. That is, multiple interfaces can provide packets to the same buffer.
- The system deletes the buffer when the session configuration is removed.
- Each node writes a file on the active RP, which contains the node ID of the node on which the buffer was located.

These stages describe how SPAN to file works when you attach multiple interfaces to a monitor session:

- Multiple interfaces are attached to a session.
- The system sends the packets to the interfaces on the same node.
- The packets are saved to the same file. Bundle interfaces can be attached to a session with a file destination, which is similar to attaching individual interfaces.

## Action commands for SPAN to file

Action commands for SPAN to file are capabilities that:

- allow you to start and stop network packet collection
- write the contents of a packet buffer to disk without stopping the packet capture for all the sessions with active packet collection
- run on sessions where the destination is a file, and
- autocomplete names of the globally configured SPAN to file sessions.

This table details the action commands for SPAN to file:

**Table 16: Action commands for SPAN to file**

| Action | Command                                                                           | Description                                                                                                                                                                  |
|--------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Start  | <code>monitor-session &lt;name&gt;</code><br><code>packet-collection start</code> | Use this command to start writing packets for the specified session to the configured buffer. This command has no effect for sessions configured as <code>always-on</code> . |

| Action | Command                                                                                                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stop   | <pre>monitor-session &lt;name&gt; packet-collection stop [ discard-data   write directory &lt;dir&gt; filename &lt;filename&gt; ]</pre> | <p>Use this command to stop writing packets to the configured buffer. If you specify the <code>discard-data</code> option, the system clears the buffer. If you specify the <code>write</code> option, the system writes the buffer to disk before clearing it.</p> <p>When writing the buffer to disk, save the file in .pcap format at the following location:<br/> <code>/&lt;directory&gt;/&lt;node_id&gt;/&lt;filename&gt;.</code><br/> If you include a .pcap extension when specifying the filename, the system will remove it to prevent the extension from being added twice.</p> <p>This command returns an error for sessions configured as <code>always-on</code>.</p>                                           |
| Write  | <pre>monitor-session &lt;name&gt; packet-collection write [directory &lt;dir&gt;] [filename &lt;filename&gt;]</pre>                     | <p>Use this command to write the contents of a packet buffer to disk without stopping the packet capture. The write command is available only for sessions with active packet collection. You can start the packet collection explicitly with the action command or through <code>always-on</code> collection.</p> <p>You may specify the full directory path and file name where the buffer needs to be written. If you specify the directory, it must already exist. If the directory or file name are not specified, the following default values are used:</p> <p>Directory: <code>/misc/scratch/SPAN/&lt;node&gt;/</code><br/> Filename:<br/> <code>&lt;session_name&gt;_&lt;node&gt;_&lt;timestamp&gt;.pcap</code></p> |

## Configure SPAN to file

Use these steps to configure SPAN to file:

### Procedure

**Step 1** Create a monitor session.

**Example:**

```
interface GigabitEthernet 0/0/0/0
monitor-session mon1 [ethernet|ipv4|ipv6|mpls-ipv4|mpls-ipv6]
```

**Step 2** Configure a **mon1** monitor session.

**Example:**

```
monitor-session mon1 ethernet
destination file size 230000
!
```

In this example, omitting the **buffer-type** option results in default circular buffer.

**Step 3** Configure a **mon2** monitor session.

**Example:**

```
monitor-session mon2 ethernet
destination file size 1000 buffer-type linear
!
```

**Step 4** Attach monitor session to a physical or bundle interface.

**Example:**

```
Router#show run interface Bundle-Ether 1
Fri Apr 24 12:12:59.348 EDT
interface Bundle-Ether1
monitor-session ms7 ethernet|ipv4|ipv6|mpls-ipv4|mpls-ipv6]
[direction {rx-only|tx-only|both[SW(1)]} [port-level]
acl [<acl_name>]!
```

**Step 5** Verify the packet collection status.

**Example:**

```
Router#show monitor-session status
Monitor-session mon1
Destination File - Packet collecting
=====
Source Interface Dir. Status

Hu0/9/0/2 Rx Operational
Monitor-session mon2
Destination File - Packet collecting
=====
Source Interface Dir Status

BE2.1. Rx Operational
```

---

If packet collection is not active, this line is displayed:

```
Monitor-session mon2
Destination File - Not collecting
```

## Configure SPAN to file for truncation

Follow these steps to configure SPAN to file for truncation.

## Procedure

- Step 1** Create a SPAN to file monitor session for mirroring the packets with truncation enabled. Use the **mirror first** command in monitor session configuration.

**Example:**

```
monitor-session <name> [ethernet]
destination file [size <kbytes>] [buffer-type linear|circular]
mirror first <number>
```

- Step 2** Attach the interfaces to the monitor session.

**Example:**

```
interface <>
 monitor-session session-name ethernet direction rx-only|tx-only|both [acl [acl_name]
```

- Step 3** Verify the configuration.

**Example:**

```
Router#show monitor-session status
Monitor-session mon1
Destination File - Packet collecting
=====
Source Interface Dir Status

Hu0/9/0/2 Rx Operational
Monitor-session mon2
Destination File - Packet collecting
=====
Source Interface Dir Status

BE1 Tx Operational
Monitor-session mon3
Destination File - Packet collecting
=====
Source Interface Dir Status

BE2.1 Both Operational
```

## Configure SPAN to file for direction

Follow these steps to configure SPAN to file for truncation.

## Procedure

- Step 1** Create a SPAN to file monitor session.

**Example:**

```
monitor-session mon2 ethernet
destination file
!
```

**Step 2** Attach the interfaces to the monitor session.

**Example:**

```
interface <>
monitor-session session-name ethernet direction rx-only|tx-only|
acl [acl_name]
```

**Step 3** Verify the configuration.

**Example:**

```
Router#show monitor-session status
Monitor-session mon1
Destination File - Packet collecting
=====
Source Interface Dir Status

Hu0/9/0/2 Rx Operational
Monitor-session mon2
Destination File - Packet collecting
=====
Source Interface Dir Status

BE1 Tx Operational
Monitor-session mon3
Destination File - Packet collecting
=====
Source Interface Dir Status

BE2.1 Both Operational
```

## Always-On SPAN to file with periodic write

The Always-On SPAN to file is a traffic mirroring feature that

- serves as a more reliable tool in investigating unexpected packet drops and traffic blackholing, and
- allows diagnosis of issues without reproducing faults, changing configurations, or needing prior user interaction before the event.

### Always-On SPAN to file functionalities

These are the key functionalities of Always-On SPAN to file:

- **Default SPAN enablement:** Enables a default SPAN to-file session for packet forwarding and buffer drops automatically, provided the platform supports it. The session is always active and periodically writes to the disk without stopping the capture, up to the maximum configured storage capacity limit. This functionality ensures continuous packet capture and storage.

You can disable this session if it's not needed using the **monitor-session default-capture-disable** command.

- SPAN truncation and sampling: Allows for packet truncation and sampling for software-mirrored packets like SPAN to file, even if hardware support isn't available. This functionality enhances flexibility by enabling these operations within the software.
- Always-on SPAN to file: Automatically starts packet capture when the file destination is configured, without requiring additional action commands. This functionality ensures immediate and continuous packet capture.

Use the **destination file always-on** command in monitor-session configuration mode to enable always-on packet capture.

- SPAN to file continuous capture: Provides the ability to write packet data to a file without stopping the ongoing packet capture. This capability ensures uninterrupted packet monitoring and data collection.

Use the **monitor-session <session name> packet-collection write** [directory <dir>] [filename <file>] command to write the current packet buffer to a file without stopping packet collection. If you don't use the optional keywords `directory` and `filename`, the system writes the buffer contents to a file named <session\_name>\_<node location>\_<timestamp>.pcap in a default capture directory.

- SPAN periodic file writing: Allows you to set a period after which the buffered SPAN to file packet data is automatically written to a file. This automatic writing prevents data loss if there is a system reload and ensures persistent storage of captured packets. The feature includes configurable limits to manage file storage effectively, ensuring user-written files remain intact and session-specific data management doesn't impact other sessions.

Use the **destination file** [size <kbytes>] [always-on [periodic-write <secs> [capacity [<num> <kB|MB|GB>]]]] command in monitor-session configuration mode to set the file writing interval.

When the periodic-write option is used, the contents of the buffer are written to a file named <session\_name>\_<node location>\_<timestamp>.pcap, in a default capture directory  
/misc/scratch/SPAN/<node>/.

## Storage capacity management and file retention rules

There are two configurable capacity options to manage the periodically written files.

- A per-session limit: The maximum storage capacity for the set of files captured periodically for an individual monitor session. When this limit is exceeded, the system automatically deletes the oldest files to make room for new ones. Depending on the newer file size, it may delete multiple older files. In cases where only a single file is captured, the file is written completely, even if its size exceeds the per-session capacity limit. The file remains stored until another file is captured for that session.

When a new file is significantly larger than the previous captures, the system may delete all existing files, leaving only the new file. This ensures that the new file is saved in its entirety without any truncation.

Use the **destination file [capacity <num> <kB|MB|GB>]** command to configure the per-session capacity limit.

- A global limit: The total storage capacity for all files captured by SPAN on disk. If this limit is exceeded, further write operations don't happen. Stopping the write operation without deleting files protects periodic and user-triggered writes in the default directory that remain unremoved or uncleared, regardless of session.

Use the **monitor-session local-capture-capacity <num> <kB|MB|GB>** command to configure the global capacity limit.




---

**Note** This capacity limit configuration applies only to files written in the default directory. Any files moved out of the default directory don't count toward this limit.

---

If you don't configure these capacity parameters, the system uses default values specific to the platform variant to manage storage capacity.

## Benefits of Always-On SPAN to file with periodic write

These are some of the benefits of the Always-On SPAN to file feature.

- **Reliable diagnostics:** Allows investigation of unexpected packet drops and traffic blackholing without the need to reproduce fault scenarios or change configurations.
- **Continuous packet capture:** Ensures uninterrupted packet monitoring and data collection by continuously capturing and storing packets without stopping.
- **Immediate activation:** Configuring the file destination automatically starts packet capture, ensuring immediate and continuous monitoring. Activating immediately reduces the risk of missing important data because of user mistakes or delays.
- **Data loss prevention:** Periodically writes packet data to a file, preventing data loss if there are system reloads and ensuring persistent storage.
- **Efficient storage management:** Provides configurable storage limits for individual sessions and overall capture, managing space effectively without deleting user-written files.
- **User control:** Allows you to disable default SPAN sessions and configure storage limits, giving the control over the packet capture and storage settings.

## Guidelines for Always-On SPAN to file with periodic write

These guidelines apply to Always-On SPAN to file with periodic write:

- The platform enables the default SPAN to file session only if it supports it.
- If you do not specify directory and filename options for the **packet-collection write** action command, the system saves buffer contents as `<session_name>_<node location>_<timestamp>.pcap` in the default capture directory `/misc/scratch/SPAN/<node>/.`
- Ensure that the user-specified directory already exists before you use it in the **packet-collection write** action command.
- The system enforces a per-session storage limit; exceeding it results in the deletion of the oldest captures.
- The system enforces a global storage limit for all captured files; exceeding this prevents further write operations.
- The system enforces the global storage limit only on the default capture directory.
- If you do not configure storage capacity limit parameters, the system uses platform-specific default values.



- When you change a SPAN-to-File session from **always-on** to **on-demand**, you must explicitly stop packet collection or write the packet buffer to enable on-demand operation.

## Configure Always-On SPAN to file with periodic write

This section includes configuration for always-on SPAN to file with periodic write and default enablement.

### Procedure

- 
- Step 1** (Optional) Disable the default SPAN to file session by using the **monitor-session default-capture-disable** command. Default SPAN to file session for packet forwarding and buffer drops is enabled automatically.
- Example:**
- ```
Router#configure
Router(config)#monitor-session default-capture-disable
Router(config)#commit
```
- Step 2** Enable Always-on SPAN to file by using the **destination file always-on** command in monitor-session configuration mode.
- Example:**
- ```
Router(config)#monitor-session test
Router(config-mon)#destination file always-on
Router(config-mon)#commit
```
- Step 3** Write the current packet buffer to a file without stopping packet collection by using the **monitor-session <session name> packet-collection write** [directory <dir>] [filename <file>] action command.
- Example:**
- ```
Router#monitor-session test packet-collection write directory var/xr/scratch/SPAN/test file testfile
```
- Step 4** Configure an interval for automatically writing buffered packet data to a file by using the **destination file** [size <kbytes>] [always-on [periodic-write]] command in monitor-session configuration mode.
- Example:**
- ```
Router(config)#monitor-session test
Router(config-mon)#destination file always-on periodic-write 300
Router(config-mon)#commit
```
- Step 5** Set the storage capacity limit for all monitor-sessions by using the **monitor-session local-capture-capacity <num> <kB|MB|GB>** command in global configuration mode.
- Example:**
- ```
Router(config)#monitor-session local-capture-capacity 300 MB
Router(config)#monitor-session test
```
- Step 6** Set the per-session limit by using the **destination file [capacity <num> <kB|MB|GB>]** command in monitor-session configuration mode, and save the changes.
- Example:**
- ```
Router(config-mon)#destination file always-on periodic-write 300 capacity 100 MB
Router(config-mon)#commit
```

**Step 7** Verify the running configuration by using the **show running-config** command.

**Example:**

```
monitor-session test ethernet
 destination file always-on periodic-write 300 capacity 100 MB
!
monitor-session local-capture-capacity 300 MB
monitor-session default-capture-disable
```

**Step 8** Verify the monitor-session details by using the **monitor-session status detail** command.

**Example:**

```
Router#show monitor-session status detail
Monitor-session test
 Destination File - Packet collecting (always-on)
 Periodic write interval: 300 seconds
 Maximum periodic capture capacity: 100MB
 Source Interfaces


```

**Step 9** Verify global configuration items and information about platform capabilities by using the **show monitor-session status internal** command.

**Example:**

```
Router#show monitor-session status internal
Global Configuration:
 Router ID: Default
 Global local-capture-capacity: 300MB
 Default session disabled
Write command supported

Information from SPAN Manager and MA on all nodes:
Monitor-session test (ID 0x00000001) (Ethernet)
SPAN Mgr: Destination File - FileID:0
 Filename/directory name not set
 Last error: Success

Information from SPAN EA on all nodes:
Monitor-session 0x00000001 (Ethernet)
0/RP0/CPU0: Name 'test', destination file FileID:0
 Filename/directory name not set
Platform, 0/RP0/CPU0:

 Monitor Session ID: 1
 Truncation Size: 0
 Buffer type: Circular
 Buffer size: 2000
```

## SPAN to file with unique capture

The SPAN to file with unique capture is a traffic mirroring feature that

- enables you to capture a single representative packet for each punt-reason or interface in a SPAN to file session,
- simplifies packet analysis

- reduces redundant data
- introduces a new buffer that ensures only one packet per port and punt reason is saved, and
- allows you to retain interesting packets while filtering out repetitive ones, making flow analysis more efficient.

SPAN to file with unique capture is implemented through the updated **monitor-session** command with options, such as **unique-punt** and **unique-port**.

### Benefits of SPAN to file with unique capture

SPAN to file with unique capture offers several key benefits:

- Retains individual packets of interest, even in the presence of many repeated packets.
- Prevents important packets from being overwhelmed by repetitive traffic.
- Captures a variety of packet types for efficient and focused analysis.

## How SPAN to file with unique capture works

### Summary

Here's how SPAN to file with unique capture works:

- The SPAN to file with unique capture captures drop packets mirrored to the SPAN drop node and saves them to a PCAPng file.
- The feature ensures uniqueness by identifying packets based on either the punt reason or the interface (port).
- The process is managed by a database or table, ensuring only one packet is captured per punt reason or port in a unique buffer, separate from the main SPAN buffer.

## Configure SPAN to file with unique capture

To configure SPAN-to-file with unique option, you must first create a monitor session using the **monitor-session** command. Choose the appropriate options to filter the drop traffic and specify whether to capture unique packets per punt reason, per port, or both.

### Procedure

**Step 1** Create a monitor session.

#### Example:

This example shows how to create a monitor-session.

```
Router(config)# monitor-session mon1 ethernet
Router(config-mon)#
```

**Step 2** Configure monitor session to capture unique packets by using the **unique-punt** and **unique-port** options available within the **drops** command.

**Example:**

This example shows how to capture unique packets for each punt reason for packet-processing drops.

```
Router(config)# monitor-session mon1 ethernet
Router(config-mon)# destination interface tunnel-ip2
Router(config-mon)# drops packet-processing rx unique-punt
Router(config)# commit
```

**Note**

If neither packet-processing nor traffic-management is explicitly configured, the SPAN session will, by default, capture drops from both traffic-management and packet-processing.

---



# CHAPTER 7

## Mirror forward-drop packets

This chapter captures and analyzes packets dropped by a router during forwarding. It helps network administrators understand blocked traffic, identify security threats, troubleshoot issues, and optimize performance by mirroring these packets to a specified destination.

- [Mirror forward-drop packets, on page 75](#)
- [Guidelines for mirroring forward-drop packets, on page 77](#)
- [Restrictions for mirroring forward-drop packets, on page 77](#)
- [Configure forward-drop packets, on page 77](#)

## Mirror forward-drop packets

Mirroring forward-drop packets is a network monitoring feature that

- captures and analyzes packets that a router drops while forwarding them
- identifies the source of potential security threats, and
- takes proactive measures to avoid escalation of the issue.

Table 17: Feature History Table

| Feature Name                   | Release Information | Description                                                                                                                         |
|--------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Mirroring forward-drop packets | Release 25.1.1      | Introduced in this release on: Fixed Systems ( 8010 [ASIC: A100])<br>This feature is now supported on Cisco 8011-4G24Y4H-I routers. |

| Feature Name                   | Release Information | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mirroring forward-drop packets | Release 24.4.1      | <p>Introduced in this release on: Fixed Systems(8200, 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*).</p> <p>This feature with the mirroring and analysis of packets dropped during the forwarding process helps identify the types of traffic that are blocked, analyze potential security threats, troubleshoot, and optimize network performance.</p> <p>*This feature is now supported on:</p> <ul style="list-style-type: none"> <li>• 8212-48FH-M</li> <li>• 8711-32FH-M</li> <li>• 8712-MOD-M</li> <li>• 88-LC1-12TH24FH-E</li> <li>• 88-LC1-52Y8H-EM</li> <li>• 88-LC1-36EH</li> </ul>                                                                                                                      |
| Mirroring forward-drop packets | Release 7.5.4       | <p>Mirroring forward-drop packets feature copies or mirrors the packets that are dropped during the forwarding process at the router ingress to a configured destination. These mirrored packets can be captured and analyzed using network monitoring tools. The analysis of dropped packets helps you understand the types of traffic that are blocked, analyze potential security threats, troubleshoot, and optimize network performance.</p> <p>This feature introduces the following changes:</p> <ul style="list-style-type: none"> <li>• <b>CLI:</b> <a href="#">drops</a></li> <li>• <b>YANG Data Model:</b> New XPath for Cisco-IOS-XR-um-monitor-session-cfg.yang (see <a href="#">GitHub</a>, <a href="#">YANG Data Models Navigator</a>)</li> </ul> |

In a network, packets are forwarded from one device to another until they reach their destination. However, in some cases, routers may drop packets during this forwarding process. These packets are known as forward-drop packets.

Packet drop can happen due to congestion on the network, errors in the packet header or payload, or blocking by firewalls or Access Control Lists (ACL). These forward-drop packets are typically discarded before they can reach their intended destination and may need to be re-transmitted by the source device.

This feature supports mirroring of these forward-drop packets at the ingress, Rx direction, to another destination. When a global forward-drop session is configured for the router, the forward-drop packets at the ingress are mirrored or copied to the configured destination. You can configure the mirror destination as a file, for SPAN to file sessions, or an IPv4 GRE tunnel ID, for ERSPAN.

## Benefits of mirroring forward-drop packets

These are the benefits of mirroring forward-drop packets to a suitable destination:

- By mirroring and analyzing forward-drop packets, network administrators gain better visibility into the types of traffic that are blocked by the firewalls and access control lists (ACL).
- As the original dropped packet is forwarded without any change, it helps in identifying the source of potential security threats.
- Analyzing forward-drop packets helps troubleshoot network issues that may be causing the packet drop. This helps in taking proactive measures to avoid escalation of the issue.

## Guidelines for mirroring forward-drop packets

These guidelines apply to mirroring forward-drop packets:

- Only one global forward-drop session can be configured on a router.
- When traffic-class is configured under monitor-session for forward-drop, the Type of Service (ToS) byte of the outgoing ERSPAN packet is overwritten with the configured traffic-class value.
- In-band traffic destined to router management interface cannot be captured using this functionality.
- For ERSPAN sessions that monitor forward-drop packets, a default value of 0 is used for the encapsulation traffic class, irrespective of the DSCP value assigned for the tunnel.

## Restrictions for mirroring forward-drop packets

These restrictions apply to mirroring forward-drop packets:

- ERSPAN counters are not updated for forward-drop packets.
- Not all packets that are dropped by NPU are mirrored.

## Configure forward-drop packets

Perform the these steps on the router to configure a global session for mirroring forward-drop packets:

### Procedure

---

**Step 1** Configure the tunnel mode.

**Example:**

```
Router(config)# interface tunnel-ip 2
Router(config-if)# tunnel mode gre ipv4
```

**Step 2** Configure the tunnel source.

**Example:**

```
Router(config-if)# tunnel source 20.20.20.20
```

**Step 3** Configure the tunnel destination.

**Example:**

```
Router(config-if)# tunnel destination 192.1.1.3
Router(config-if)# exit
```

**Step 4** Configure a traffic mirroring session.

**Example:**

```
Router(config)# monitor-session mon2 ethernet
```

**Step 5** Associate a destination interface with the traffic mirroring session.

**Example:**

```
Router(config)# destination interface tunnel-ip2
```

**Step 6** Run **drops** command to start mirroring forward-drop packets.

**Example:**

```
Router(config)# drops packet-processing rx
Router(config)# commit
```

**Step 7** Verify the forward-drop packets are mirrored using the **show monitor-session** command.

**Example:**

```
Router# show monitor-session mon2 status detail
Mon Aug 15 19:14:31.975 UTC
Monitor-session mon2
 Destination interface tunnel-ip2
 All forwarding drops:
 Direction: Rx
 Source Interfaces

```

---





## CHAPTER 8

# Mirror buffer drop packets

This chapter details the mirror buffer drop packets feature, which captures and mirrors packets dropped by a router's Traffic Management (TM) buffer due to overflow, specifically exact or statistical metering. It enables analysis of these dropped packets, offering benefits like data preservation, control plane stability, and failover assurance.

- [Mirror buffer drop packets, on page 79](#)
- [Guidelines for mirroring buffer drop packets, on page 81](#)
- [Configure buffer drop packets mirroring for SPAN to file, on page 81](#)
- [Configure buffer drop packets mirroring for ERSPAN, on page 83](#)

## Mirror buffer drop packets

Mirroring buffer drop packets is a network monitoring feature that

- captures packets dropped by a router due to buffer overflow, and
- sends these packets to a monitoring system for analysis.

**Table 18: Feature History Table**

| Feature Name                  | Release Information | Description                                                                                                                                                                                                              |
|-------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mirroring buffer drop packets | Release 25.4.1      | Introduced in this release on: Fixed Systems (8010 [ASIC: A100])(select variants only)<br><br>*This feature is supported on: <ul style="list-style-type: none"><li>• 8011-12G12X4Y-A</li><li>• 8011-12G12X4Y-D</li></ul> |
| Mirroring buffer drop packets | Release 25.1.1      | Introduced in this release on: Fixed Systems ( 8010 [ASIC: A100])<br><br>This feature is now supported on: <ul style="list-style-type: none"><li>• 8011-4G24Y4H-I</li></ul>                                              |

| Feature Name                  | Release Information | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mirroring buffer drop packets | Release 24.4.1      | <p>Introduced in this release on: Fixed Systems(8200, 8700)(select variants only*); Modular Systems (8800 [LC ASIC: P100])(select variants only*).</p> <p>This feature which mirrors packets dropped by the Traffic Management (TM) buffer when it is full and starts dropping incoming packets so that the mirrored copy of the dropped packets can be retained and stored is now supported on the following hardware.</p> <p>*This feature is now supported on:</p> <ul style="list-style-type: none"> <li>• 8212-48FH-M</li> <li>• 8711-32FH-M</li> <li>• 8712-MOD-M</li> <li>• 88-LC1-12TH24FH-E</li> <li>• 88-LC1-52Y8H-EM</li> <li>• 88-LC1-36EH</li> </ul>                                                                                                                                                               |
| Mirroring buffer drop packets | Release 24.2.11     | <p>The SPAN-to-file and ERSPAN mirroring capability is enhanced to mirror dropped packets by the Traffic Management (TM) buffer when it's full and starts dropping incoming packets. This capability allows you to retain and store a mirrored copy of the dropped packets, and work effectively even during process restarts or network failovers, providing a dependable solution for traffic monitoring.</p> <p>This feature is supported only on Cisco Silicon One P100- and Q200-based routers.</p> <p>This feature introduces the following changes:</p> <ul style="list-style-type: none"> <li>• <b>CLI:</b> <a href="#">drops</a></li> <li>• <b>YANG Data Model:</b> New XPath for <code>Cisco-IOS-XR-Ethernet-SPAN-cfg.yang</code> (see <a href="#">GitHub</a>, <a href="#">YANG Data Models Navigator</a>)</li> </ul> |

### Packet drop scenarios

Traffic Management (TM) buffer drops typically occur due to network congestion. The TM buffer temporarily stores packets for processing, but if the buffer becomes full, it cannot accommodate additional packets and starts discarding or dropping them. This situation arises when incoming packets arrive faster than the buffer can process them.

With enhanced SPAN to file and ERSPAN mirroring capabilities, the router retains a mirrored copy of these dropped packets for analysis.

This feature covers the following packet drop scenarios:

- **Exact Meter Drops:** When the incoming traffic on an interface exceeds the rate enforced by a configured policer, the router drops excess packets. These drops are reported as `PACKET_GOT_DROPPED_DUE_TO_EXACT_METER` (displayed as `TM_EXACT_METER_DROP`).
- **Statistical Meter Drops or Mirror Buffer Drop Packets:** When traffic is sent at 100% line rate to an ingress interface under a regular SPAN-to-File configuration, the router may drop packets due to statistical metering. These drops are reported as `PACKET_GOT_DROPPED_DUE_TO_STATISTICAL_METER` (Displayed as `TM_STATISTICAL_METER_DROP`).

## Benefits of mirroring buffer drop packets

These are the benefits of mirroring buffer drop packets:

- **Data Preservation:** During process restarts, while new packet logging is paused, all previously collected data remains intact, ensuring no critical diagnostic information is lost.
- **Control Plane Stability:** The operation of this feature remains unaffected by restarts of any control plane process, allowing for uninterrupted traffic analysis.
- **Failover Assurance:** For interfaces not located on the Route Processor (RP), this feature ensures a seamless failover experience. Traffic mirroring and packet capture proceed without disruption, even amidst hardware changes or network reconfigurations, safeguarding continuous network analysis.

## Guidelines for mirroring buffer drop packets

These guidelines apply to mirroring buffer drop packets:

- SPAN to File Buffer drop is a global configuration, not specific to any interface.
- Mirroring of buffer drop packets is supported only on the ingress interface.
- Only the packets dropped by the reasons `TM_EXACT_METER_DROP` and `TM_STATISTICAL_METER_DROP` are mirrored.
- Maximum of one drop session with destination as Span to File and one drop session with destination as ERSPAN can be configured globally per router.
- From Release 24.2.11, one forward-drop session and one TM buffer drop session are supported for file and GRE tunnel interface destinations.
- For ERSPAN sessions that monitor buffer drop packets, a default value of 0 is used for the encapsulation traffic class, irrespective of the DSCP value assigned for the tunnel.
- ERSPAN counters are not updated for buffer drop packets.

## Configure buffer drop packets mirroring for SPAN to file

Perform the these steps on the router to configure a global session for mirroring buffer drop packets for SPAN to file destination:

## Procedure

**Step 1** Configure a traffic mirroring session and specify the destination as file.

**Example:**

```
Router(config)# monitor-session S2F_sessnethernet
Router(config-mon)# destination file
Router(config-mon)# exit
```

**Step 2** Attach SPAN to file monitor session to the interface.

**Example:**

```
Router(config)# interface HundredGigE0/0/0/0
Router(config-if)# monitor-session S2F_sessnethernet direction rx-only
Router(config-if-mon)# exit
```

**Step 3** Create a global monitor session for TM drop packets and enable the TM buffer drop feature for SPAN to file destination.

**Example:**

```
Router(config)# monitor-session mon1 ethernet
Router(config-mon)# destination file
Router(config-mon)# drops traffic-management rx
Router(config-mon)# commit
```

**Step 4** Configure a policer on the ingress interface and send the traffic at a line rate faster than the policer. The router drops the packets with the reason PACKET\_GOT\_DROPPED\_DUE\_TO\_EXACT\_METER (displayed as TM\_EXACT\_METER\_DROP).

**Example:**

```
Router(config)# class-map match-any dscpl
Router(config-cmap)# match dscp ipv4 1
Router(config-cmap)# end-class-map
Router(config)# policy-map test-police-1R2C
Router(config-pmap)# class dscpl
Router(config-pmap-c)# police rate 100 mbps
Router(config-pmap-c-police)# exit
Router(config-pmap-c)# class class-default
Router(config-pmap-c)# police rate 100 mbps
Router(config-pmap-c-police)# exit
Router(config-pmap-c)# end-policy-map
Router(config)# interface HundredGigE0/0/0/0
Router(config-if)# service-policy input test-police-1R2C
Router(config-if)# commit
```

**Step 5** Configure a regular SPAN to file on the ingress interface and send the traffic at 100% line rate. The router drops the packets with the reason PACKET\_GOT\_DROPPED\_DUE\_TO\_STATISTICAL\_METER (Displayed as: TM\_STATISTICAL\_METER\_DROP).

**Step 6** Verify the buffer drop packets are mirrored using the **show monitor-session status** command.

**Example:**

```
Router# show controllers npu stats traps-all instance all location 0/RP0/CPU0
Trap Type NPU Trap Punt Punt Punt Punt Configured Hardware
Policer Avg-Pkt Packets Packets
ID ID Dest VoQ VLAN TC Rate (pps) Rate (pps)
```

| Level                            | Size | Accepted     | Dropped  |     |          |     |      |   |     |     |
|----------------------------------|------|--------------|----------|-----|----------|-----|------|---|-----|-----|
| <b>TM_EXACT_METER_DROP</b>       |      |              | 0        | 256 | RPLC_CPU | 208 | 1538 | 0 | 542 | 523 |
| NPU                              | N/A  | <b>25</b>    | 18167604 |     |          |     |      |   |     |     |
| <b>TM_STATISTICAL_METER_DROP</b> |      |              | 0        | 257 | RPLC_CPU | 208 | 1538 | 0 | 542 | 523 |
| NPU                              | N/A  | <b>31143</b> | 18167604 |     |          |     |      |   |     |     |

```
Router# show spp node-counters location all | i SPAN
SPAN to File: 55724
SPAN drop: 31168
```

The Packets Accepted counter of the statistical meter dropped packets and exact meter dropped packets in **show controllers npu stats traps-all** should match the "SPAN drop" counter in **show spp node-counters** within an acceptable range. This is because the statistics displayed are not updated in real-time; they refresh every 30 seconds from the hardware.

## Configure buffer drop packets mirroring for ERSPAN

Perform these steps on the router to configure a global session for mirroring buffer drop packets for SPAN to file destination:

### Procedure

**Step 1** Configure the tunnel mode.

**Example:**

```
Router(config)# interface tunnel-ip2
Router(config-if)# tunnel mode gre ipv4
```

**Step 2** Configure the tunnel source.

**Example:**

```
Router(config-if)# tunnel source 10.10.10.10
```

**Step 3** Configure the tunnel destination.

**Example:**

```
Router(config-if)# tunnel destination 192.0.2.1
Router(config-if) exit
```

**Step 4** Configure a traffic mirroring session.

**Example:**

```
Router(config)# monitor-session mon2 ethernet
```

**Step 5** Associate a destination interface with the traffic mirroring session.

**Example:**

```
Router(config-mon)# destination interface tunnel-ip2
```

**Step 6** Configure the TM buffer drop session and save the changes.

**Example:**

```
Router(config-mon)# drops traffic-management rx
Router(config-mon)# commit
```

**Step 7** Configure a policer on the ingress interface and send the traffic at a line rate faster than the policer. The router drops the packets with the reason PACKET\_GOT\_DROPPED\_DUE\_TO\_EXACT\_METER (displayed as TM\_EXACT\_METER\_DROP).

**Example:**

```
Router(config)# class-map match-any dscp1
Router(config-cmap)# match dscp ipv4 1
Router(config-cmap)# end-class-map
Router(config)# policy-map test-police-1R2C
Router(config-pmap)# class dscp1
Router(config-pmap-c)# police rate 100 mbps
Router(config-pmap-c-police)# exit
Router(config-pmap-c)# class class-default
Router(config-pmap-c)# police rate 100 mbps
Router(config-pmap-c-police)# exit
Router(config-pmap-c)# end-policy-map
Router(config)# interface HundredGigE0/0/0/0
Router(config-if)# service-policy input test-police-1R2C
Router(config-if)# commit
```

**Step 8** Configure a regular SPAN to file on the ingress interface and send the traffic at 100% line rate. The router drops the packets with the reason PACKET\_GOT\_DROPPED\_DUE\_TO\_STATISTICAL\_METER (Displayed as: TM\_STATISTICAL\_METER\_DROP).

**Step 9** Verify the buffer drop packets are mirrored using the **show monitor-session status** command.

**Example:**

```
Router# show controllers npu stats traps-all instance all location 0/RP0/CPU0
```

| Trap Type                        |         | NPU      | Trap     | Punt | Punt | Punt     | Punt | Configured | Hardware   |
|----------------------------------|---------|----------|----------|------|------|----------|------|------------|------------|
| Policer                          | Avg-Pkt | Packets  | Packets  |      |      |          |      |            |            |
| Level                            | Size    | Accepted | ID       | ID   | Dest | VoQ      | VLAN | TC         | Rate (pps) |
|                                  |         |          | Dropped  |      |      |          |      |            |            |
| <b>TM_EXACT_METER_DROP</b>       |         |          |          |      |      |          |      |            |            |
| NPU                              | N/A     | 25       | 18167604 | 0    | 256  | RPLC_CPU | 208  | 1538       | 542        |
| <b>TM_STATISTICAL_METER_DROP</b> |         |          |          |      |      |          |      |            |            |
| NPU                              | N/A     | 31143    | 18167604 | 0    | 257  | RPLC_CPU | 208  | 1538       | 542        |

```
Router# show spp node-counters location all | i SPAN
SPAN to File: 55724
SPAN drop: 31168
```

The Packets Accepted counter of the statistical meter dropped packets and exact meter dropped packets in **show controllers npu stats traps-all** should match the SPAN drop counter in **show spp node-counters** within an acceptable range. This is because the statistics displayed are not updated in real-time; they refresh every 30 seconds from the hardware.



## CHAPTER 9

# File Mirroring

This chapter details the File mirroring feature, which automatically copies files and directories from an active Route Processor (RP) to a standby RP or RSP. It eliminates manual intervention for file synchronization, ensuring redundancy.

- [File mirroring, on page 85](#)
- [Restrictions for file mirroring, on page 86](#)
- [Configure file mirroring, on page 86](#)

## File mirroring

File mirroring is a file management feature that

- enables the router to copy files or directories automatically from `/hddisk:/mirror` location in active RP to `/hddisk:/mirror` location in standby RP or RSP without user intervention or EEM scripts, and
- enables the feature by using **mirror enable** and **mirror enable checksum** options.

**Table 19: Feature History Table**

| Feature Name   | Release Information | Description                                                                                                                                                                       |
|----------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File mirroring | Release 7.0.14      | This feature enables the router to copy files and directories automatically from an active RP to a standby RP thus eliminating the manual intervention or the use of EEM scripts. |

You can use these commands to enable the file mirroring feature:

- **mirror enable** - The `/hddisk:/mirror` directory is created by default, but file mirroring functionality is only enabled by executing the `mirror enable` command from configuration terminal. Status of the mirrored files can be viewed with `show mirror status` command.
- **mirror enable checksum** - The **mirror enable checksum** command enables MD5 checksum across active to standby RP to check integrity of the files. This command is optional.

## Restrictions for file mirroring

These restrictions apply to file mirroring:

- Supported only on Dual RP systems.
- Supports syncing only from active to standby RP. If files are copied into standby `/harddisk:/mirror` location, it won't be synced to active RP.
- A slight delay is observed in `show mirror` command output when mirror checksum configuration is enabled.
- Not supported on multichassis systems.

## Configure file mirroring

Use these steps to configure file mirroring:

### Before you begin

File mirroring must be enabled explicitly on the router. It is not enabled by default.

### Procedure

**Step 1** Ensure mirroring is enabled.

#### Example:

```
Router#show run mirror
```

```
Thu Jun 25 10:12:17.303 UTC
mirror enable
mirror checksum
```

**Step 2** Copy running configuration to `harddisk:/mirror` location.

#### Example:

```
Router#copy running-config harddisk:/mirror/run_config
Wed Jul 8 10:25:51.064 PDT
Destination file name (control-c to abort): [/mirror/run_config]?
Building configuration..
32691 lines built in 2 seconds (16345)lines/sec
[OK]
```

**Step 3** Verify the syncing of file copied to mirror directory, use the `show mirror` command.

#### Example:

```
Router#show mirror
Wed Jul 8 10:31:21.644 PDT
% Mirror rsync is using checksum, this show command may take several minutes if you have many files.
Use Ctrl+C to abort
MIRROR DIR: /harddisk:/mirror/
% Last sync of this dir ended at Wed Jul 8 10:31:11 2020
```



```

Location |Mirrored |MD5 Checksum |Modification Time

run_config |yes |76fc1b906bec4fe08ecda0c93f6c7815 |Wed Jul 8 10:25:56 2020

```

**Table 20: Identify issues**

| If..                                           | Then..                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the checksum is disabled                       | <p>use the <b>show mirror</b> command to display the following output:</p> <pre> Router#<b>show mirror</b> Wed Jul  8 10:39:09.646 PDT MIRROR DIR: /harddisk:/mirror/ % Last sync of this dir ended at Wed Jul  8 10:31:11 2020 Location       Mirrored  Modification Time ----- run_config  yes       Wed Jul  8 10:25:56 2020 </pre>        |
| there is a mismatch during the syncing process | <p>use <b>show mirror mismatch</b> command to verify:</p> <pre> Router# <b>show mirror mismatch</b> Wed Jul  8 10:31:21.644 PDT MIRROR DIR: /harddisk:/mirror/ % Last sync of this dir ended at Wed Jul  8 10:31:11 2020 Location       Mismatch Reason       Action Needed ----- test.txt       newly created item.   send to standby </pre> |





## CHAPTER 10

# How to troubleshoot traffic mirroring

This chapter covers how to troubleshoot traffic mirroring issues. It outlines a systematic approach starting with the **show monitor-session status** command to diagnose configuration errors and operational problems. The document details various session and source interface status messages, their explanations, and recommended actions

- [Troubleshoot traffic mirroring issues, on page 89](#)
- [Additional debug commands, on page 91](#)

## Troubleshoot traffic mirroring issues

When you encounter any issue with traffic mirroring, begin troubleshooting by checking the output of the **show monitor-session status** command.

### Procedure

- Step 1** Use the **show monitor-session status** command when you encounter any issue with traffic mirroring. This command displays the recorded state of all sessions and source interfaces:

**Example:**

```
Monitor-session sess1
<Session status>
=====
Source Interface Dir Status

Gi0/0/0/0 Both <Source interface status>
Gi0/0/0/2 Both <Source interface status>
```

The **<Session status>** indicates one of these configuration errors:

| Session Status                     | Explanation                                                                                                                                                |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Session is not configured globally | The session does not exist in global configuration. Check <b>show run</b> command output to ensure that a session with a correct name has been configured. |

| Session Status                                 | Explanation                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Destination interface <intf> is not configured | The interface that has been configured as the destination does not exist. For example, the destination interface may be configured as a VLAN subinterface, but the VLAN subinterface may not yet be created.                                                                                                                   |
| Destination interface <intf> (<down-state>)    | <p>The destination interface is not in Up state in the Interface Manager. You can verify the state using the <b>show interfaces</b> command.</p> <p>Check the configuration to see what might be keeping the interface from coming up. For example, a sub-interface needs to have an appropriate encapsulation configured.</p> |

<Source interface status> reports these messages:

| Source Interface Status                              | Explanation                                                                                                                                                                                                                                                                             |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operational                                          | Everything appears to be working correctly in traffic mirroring PI. Please follow up with the platform teams in the first instance if mirroring is not operating as expected.                                                                                                           |
| Not operational (Session is not configured globally) | The session does not exist in global configuration. Check the <b>show run</b> command output to ensure that a session with the right name has been configured.                                                                                                                          |
| Not operational (destination interface not known)    | The session exists, but it either does not have a destination interface specified, or the destination interface named for the session does not exist (for example, if the destination is a sub-interface that has not been created).                                                    |
| Not operational (source same as destination)         | The session exists, but the destination and source are the same interface, so traffic mirroring does not work.                                                                                                                                                                          |
| Not operational (destination not active)             | The destination interface or pseudowire is not in the Up state. See the corresponding <i>Session status</i> error messages for suggested resolution.                                                                                                                                    |
| Not operational (source state <down-state>)          | The source interface is not in the Up state. You can verify the state using the <b>show interfaces</b> command. Check the configuration to see what might be keeping the interface from coming up (for example, a sub-interface needs to have an appropriate encapsulation configured). |
| Error: see detailed output for explanation           | Traffic mirroring has encountered an error. Run the <b>show monitor-session status detail</b> command to display more information.                                                                                                                                                      |

- Step 2** Use the **show monitor-session status detail** command to display full details of the configuration parameters, and of any errors encountered.

**Example:**

```
Router#show monitor-session status detail
Monitor-session sess1
Destination interface is not configured
Source Interfaces

HundredGigE0/0/0/0
Direction: Both
ACL match: Enabled
Portion: Full packet
Status: Not operational (destination interface not known)
HundredGigE0/0/0/2
Direction: Both
ACL match: Disabled
Portion: First 100 bytes
Status: Not operational (destination interface not known). Error: 'Viking SPAN PD' detected the
'warning' condition 'PRM connection creation failure'.
Monitor-session foo
Destination next-hop HundredGigE 0/0/0/0
Source Interfaces

HundredGigE 0/1/0/0.100:
Direction: Both
Status: Operating
HundredGigE 0/2/0/0.200:
Direction: Tx
Status: Error: <blah>

Monitor session bar
No destination configured
Source Interfaces

HundredGigE 0/3/0/0.100:
Direction: Rx
Status: Not operational(no destination)
```

### What to do next

After configuring an Ethernet interface, you can configure individual VLAN subinterfaces on the Ethernet interface.

For information about modifying Ethernet management interfaces for the shelf controller (SC), route processor (RP), and distributed RP, see the Advanced Configuration and Modification of the Management Ethernet Interface later in this document.

For information about IPv6, see the Implementing Access Lists and Prefix Lists on Cisco IOS XR Software module in the Cisco IOS XR IP Addresses and Services Configuration Guide.

## Additional debug commands

This table details the various problem scenarios and how to troubleshoot them.

| When..                                            | Then..                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| there is no route to the destination IPv4 address | <p>use the <b>status</b> command</p> <pre>Router#show monitor-session mon2 status internal Wed Oct 9 19:24:06.084 UTC Information from SPAN Manager and MA on all nodes: Monitor-session mon2 (ID 0x00000001) (Ethernet) SPAN Mgr: Destination interface tunnel-ip2 (0x0f000034) (down) Last error: Success Tunnel data: Mode: GREoIPv4 Source IP: 2.2.2.2 Dest IP: 130.10.10.2 VRF: ToS: 0 (copied) TTL: 255 DFbit: Not set 0/1/CPU0: Destination interface is not configured Tunnel data: Mode: GREoIPv4 Source IP: 2.2.2.2 Dest IP: 130.10.10.2 VRF: ToS: 0 (copied) TTL: 255 DFbit: Not set</pre> |
| there is a route to the destination IPv4 address  | <p>use this command:</p> <pre>Router#show cef ipv4 130.10.10.2 Wed Oct 9 19:25:12.282 UTC 0.0.0.0/0, version 0, proxy default, default route handler, drop adjacency, internal 0x1001011 0x0 (ptr 0x8e88d2b8) [1], 0x0 (0x8ea4d0a8), 0x0 (0x0) Updated Oct 9 19:03:36.068 Prefix Len 0, traffic index 0, precedence n/a, priority 15 via 0.0.0.0/32, 3 dependencies, weight 0, class 0 [flags 0x0] path-idx 0 NHID 0x0 [0x8e2db240 0x0] next hop 0.0.0.0/32 drop adjacency</pre>                                                                                                                      |
| a route is present                                | <p>use this command:</p> <pre>Router#show cef ipv4 130.10.10.2 Wed Oct 9 19:25:12.282 UTC 0.0.0.0/0, version 0, proxy default, default route handler, drop adjacency, internal 0x1001011 0x0 (ptr 0x8e88d2b8) [1], 0x0 (0x8ea4d0a8), 0x0 (0x0) Updated Oct 9 19:03:36.068 Prefix Len 0, traffic index 0, precedence n/a, priority 15 via 0.0.0.0/32, 3 dependencies, weight 0, class 0 [flags 0x0] path-idx 0 NHID 0x0 [0x8e2db240 0x0] next hop 0.0.0.0/32 drop adjacency</pre>                                                                                                                      |

| When..                                                                                        | Then..                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ARP for the next hop is resolved, which is done by invoking a ping command to the destination | <p>the show command output displays this output:</p> <pre> RP/0/RP0/CPU0:Router1#show monitor-session mon2 status internal Wed Oct 9 19:32:24.856 UTC Information from SPAN Manager and MA on all nodes: Monitor-session mon2 (ID 0x00000001) (Ethernet) SPAN Mgr: Destination interface tunnel-ip2 (0x0f000034) Last error: Success Tunnel data: Mode: GREoIPv4 Source IP: 2.2.2.2 Dest IP: 130.10.10.2 VRF: ToS: 0 (copied) TTL: 255 DFbit: Not set O/1/CPU0: Destination interface tunnel-ip2 (0x0f000034) Tunnel data: Mode: GREoIPv4 Source IP: 2.2.2.2 Dest IP: 130.10.10.2 VRF: ToS: 0 (copied) TTL: 255 DFbit: Not set Information from SPAN EA on all nodes: Monitor-session 0x00000001 (Ethernet) O/1/CPU0: Name 'mon2', destination interface tunnel-ip2 (0x0f000034) Platform, O/1/CPU0: Monitor Session ID: 1 Monitor Session Packets: 0 Monitor Session Bytes: 0 O/2/CPU0: Name 'mon2', destination interface tunnel-ip2 (0x0f000034) Platform, O/2/CPU0: Monitor Session ID: 1 Monitor Session Packets: 0 Monitor Session Bytes: 0 </pre> |

These are the additional trace and debug commands for traffic mirroring.

- **show monitor-session platform trace**
- **show monitor-session trace**
- **debug monitor-session platform**
- **debug monitor-session platform all**
- **debug monitor-session platform event**
- **debug monitor-session platform info**
- **show monitor-session status**
- **show monitor-session status**
- **show monitor-session status errors**
- **show monitor-session status internal**

